A New Interferometric Orbit and Fundamental Parameters for the Massive Binary Spica

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ABSTRACT

One measure of the internal structure of a massive star is provided by the observed apsidal motion constant,$\log k_{2obs}$, constrained most often in eclipsing eccentric double-line spectroscopic binaries which show an advance in the longitude of periastron. For non-eclipsing systems, a visual orbit must be established to constrain the system's inclincation. To this end, long-baseline interferometric observations of the massive, 4.01-day period, double-line spectroscopic binary Spica have been obtained with the Center for High Angular Resolution Astronomy (CHARA) Array in three-, five-, and six-telescope configurations with the Michigan Infrared Combiner (MIRC) at H-band and in two-telescope configurations with the Fiber-Linked Unit for Optical Recombination (FLUOR)at K-band. The MIRC data provide the first closure phase measurements of Spica. These data, together with two-telescope data from the Sydney University Stellar Interferometer (SUSI) at 700 nm, archival radial velocity data spanning 110 years, and archival absolute spectrophotometry from the UV to near-IR, are simultaneously fit by a detailed atmospheric model for the system, which includes the effects of tidal and rotational distortion, and constrains the orbital and stellar parameters for Spica, including its apsidal period. With the exception of the apsidal period, our best-fit parameters are largely consistant with pioneering study by Herbison-Evans et al. (1971) using the Narrabri Stellar Intensity Interferometer. Our best fit value for apsidal period, 110±? years, together with other parameters, yields an apsidal structure constant, $\log k_{2obs} = -2.2\pm?$, 0.4 dex higher than previously determined and consistant with theoretical published predictions for Spica. To check our model-fitting procedures for Spica, we have also fit CHARA/MIRC three-telescope data, archival radial velocity, and archival

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spectrophotometry for the 10.2-day, double-line spectroscopic binary ι Peg and present orbital and stellar parameters that provide an unambigous determination of the sense of orbital motion, orbital inclination, and longitude of the ascending node.

Subject headings: binaries: close — binaries: spectroscopic — methods: data analysis — methods: numerical — stars:interiors — stars: fundamental parameters (masses, radii) — stars:individual (Spica) — techniques: interferometric

1. Introduction

An ear of wheat in Virgo's left hand, the bright star Spica (α Vir = HR 5056 = HD 116658 = HIP 65474) was used by Hipparchus in the 2nd century B.C.E to measure the precession rate of the equinoxes and has been recognized as a spectroscopic binary since late 19th Century (Vogel 1891; Maury & Pickering 1897). Spica was the second star, after Capella (Anderson 1920; Merrill 1922), to have its visual binary orbital elements determined obtained by optical interferometry (Herbison-Evans et al. 1971). These data constrained the individual masses of the component stars of the then non-eclipsing system. Recent space-based photometry (Desmet et al. 2009) shows evidence for a grazing eclipse. The Spica system has an orbital period of 4.01 days and consists of two early B-type stars (Levato 1975, (B1V, B1III-IV, B2V) + B2V). The eccentric orbit ($e \simeq 0.1$) and the stars' tidal and rotational distortions lead to the advance of its line of apsides; one rotation of the line of apsides takes $\sim$100 years (Luyten & Ebbighausen 1935; Struve et al. 1958; Dukes 1974; Lyubimkov et al. 1995). Most importantly, the rate of the advance of periastron depends on the distribution of mass within the stellar components (Batten 1973, see chapter 6), mainly the larger, more massive primary star and therefore provides a method by which to constrain the internal structure of the primary star independent from the methods of asteroseismology. The orbital solution of Herbison-Evans et al. (1971) sparked several theoretical investigations regarding internal structure of the primary star (Mathis & Odell 1973; Odell 1974), however the orbital and stellar parameters from the solution of Herbison-Evans et al. (1971) indicated a more centrally concentrated mass distribution for the primary star than predicted. This disagreement was true not only for Spica but for many eccentric double-line spectroscopic binary systems which are eclipsing and have accurate radii. Over the next two decades, improvements in stellar interior models (better opacities, including core overshooting and rotation effects) significantly reduced the discrepancies between theory and observation for most systems (Claret & Gimenez 1993), but not for Spica. Claret & Willems (2002) and Claret (2003) concluded that Spica remained one of the massive binaries for which there is still significant disagreement between observation and theory regarding the apsidal constant. To better understand this discrepancy and more tightly constrain Spica's fundamental parameters, we have obtained long-baseline interferometric observations of Spica with the Center for High Angular Resolution Astronomy (CHARA) Array (ten Brummelaar et al. 2005) in three-, five-, and six-telescope configurations with the Michigan Infrared Combiner (MIRC, see Monnier et al. (2006)) at H-band, two-telescope data at CHARA with the Fiber-Linked Unit for Optical Recombination (FLUOR, see Coudé du Foresto et al. (2003)) at K-band, and two-telescope data with the Sydney University Stellar Interferometer (SUSI, see Davis et al. (1999)) at 700 nm. To aid in the analysis of these data, we developed a code to predict interferometric, spectroscopic, and photometric observables based on detailed model atmospheres for the stellar components that include the effects of tidal distortion and rotational distortion. This paper reports this analysis and a new orbital solution for the Spica system which yields an apsidal constant consistent with earlier internal structure calculations. Section 2 says... etc.

2. Interferometric Observations

We have obtained a new orbital solution for the Spica system by fitting observations from two long-baseline optical interferometry arrays CHARA and SUSI. Figure 1 shows the combined (u, v)-plane sampling of these interferometric data sets. The vast majority of interferometric data are from CHARA/MIRC. We describe these observations and data sets in below.

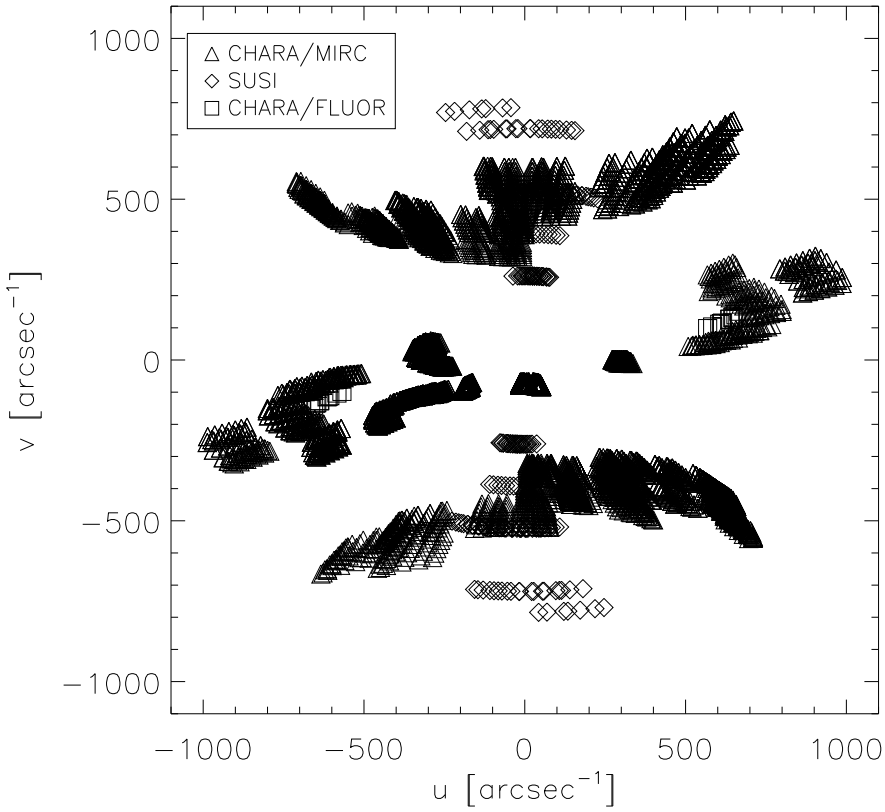

Fig. 1.— The non-contemporaneous spatial frequency coverage of Spica from CHARA/MIRC (triangles), and SUSI (diamonds)from CHARA/FLUOR (squares). These data can be found in Tables 8, 7,n and6

2.1. CHARA/MIRC Observations

To achieve the maximum possible spatial frequency and temporal converge over 4.01-day orbital period, Spica was observed on four consecutive nights with up to six telescopes of the CHARA Array and MIRC instrument on three observing runs. In May 2007 some of us (JDM and MZ) obtained 144 sets of closure phase and triple amplitude data with CHARA/MIRC with the three-telescope configuration S1-E2-W1. In June 2012 some of us (JDM, FB, NT, and EP) further enhanced the spatial frequency coverage on Spica and obtained 544 sets of closure phase and triple amplitude using five telescopes from the Array: S1,S2,E2,W1,W2. In April 2015 some of us (JPA, GS, CF, and OM) obtained 1525 sets of closure phase and triple amplitude data with both five- and six-telescope configurations (with and without the E1 telescope). Together these three runs provide 2213 sets of closure phases and triple amplitudes in up to 8 spectral channels spanning the H-band ($\lambda = 1.50 \quad 1.74\mu$m). These data probably need to be in some big machine-readable table, see Table 8 (now under construction).

2.2. CHARA/FLUOR Observations

In 2006, while MIRC was still in the commissioning phase, some of us (JPA, MJI, and AM) obtained 11 squared visibility measurements with CHARA/FLUOR in the K-band with projected east-west baselines between 262 m and 310 m. These data are listed in Table 6.

2.3. SUSI

In May and June of 2004 and in March 2005, some of us (MJI and APJ) obtained 108 squared visibility measurements of Spica with SUSI at a wavelength of 700 nm with projected north-south baselines between 37 m and 113 m. These data are listed in Table 7.

3. Archival Data

To futher constain the orbital solution, we have simulataneously fit the interferometric data described above together with archival radial velocity data spanning over 100 years and archival absolute spectrophotometric data from the ultraviolet to the near infrared. This section describes these data sets.

3.1. Archival Radial Velocities

We obtained observation dates, times, and radial velocities for Spica A and Spica B from the literature (Vogel 1890; Baker 1910; Struve & Ebbighausen 1934; Struve et al. 1958; Shobbrook et al. 1972; Riddle 2000) and coverted these data to a common format: Reduced Julian Date (RJD = JD - 2400000) and radial velocity in kilometers per second, see Table **??** (table under construction). For the Vogel (1890) dataset care was taken to convert mean Potsdam time to Universal Time (UT), before converting to RJD. The radial velocities were converted from German State Miles per second to kilometers per second. For the Baker (1910) dataset care was taken to convert Greenwich Mean Time (before 1925, GMT 0h at 12h UT) to UT. We use the uncertainties in the radial velocity measurements provided by the authors: Vogel (1890) A only: ±7.5 km/s; Baker (1910) A: ±9.9 km/s; B: ±10.2 km/s; Struve & Ebbighausen (1934) both stars ± 10 km/s; Struve et al. (1958) both stars ±4.5 km/s; Shobbrook et al. (1972) A: ±2.7 km/s, B: ±5.0 km/s; Riddle (2000) both stars ±1 km/s. Based on our model fits (see below), in many cases the radial velocity uncertainties for Spica B appear to be much larger than stated by these authors when Spica A is in conjuction with Spica B, that is when the primary and secondary spectra are heavily blended.

3.2. Archival Spectrophotometry

To future constrain the spectral energy distribution (SED) of Spica, apart from flux ratio at H-band (CHARA/MIRC), K-band (CHARA/FLUOR), and 700 nm (SUSI), we have extracted ultraviolet, optical, and near-IR absolute spectrophotometry from the literature. The International Ultraviolet Explorer (IUE)

data (λ = 115 nm to 310 nm) from the IUE Newly Extracted Spectra (INES) database (see e.g., González-Riestra et al. 2000)) and consist of two rebinned high-disperson spectra (SWP33082RL and LWP12841RL) because Spica was too bright for the low-dispersion mode of IUE. The thirteen-color photometry of (Johnson & Mitchell 1975, λ = 316 nm to 1050 nm) contrains the model SED into the near-IR.

4. A model for Spica

In order to include the effects of tidal and rotational distortion on the brightness distributions of Spica's stellar components and to gauge the significance of these effects on the synthetic interferomeric and photometric observables, we have developed a code for computing synthetic images, spectra, and photometry for the rapidly rotating and tidally distorted Roche surfaces. Although such a model is not self-consistent with regard to stellar structure (the Roche model assumes a point-mass), it is much more physically realistic than the uniform circular disk models employed by Herbison-Evans et al. (1971) A detailed description of the code is found in Appendix A.

4.1. Best Fit Orbital Parameters for Spica

As of 11 August 2015, Table 2 lists the best fit orbital and stellar parameters for the Spica system based on a simultaneous fit to the CHARA/MIRC (see Figure 12), SUSI (see Figure 3), CHARA/FLUOR (see Figure 4) interferometric data, archival radial velocities (see Figure 5) 13-color photometry (see Figure 6) Exoplorer (IUE) spectrophotometry (see Figure 7). I'm working on error bars (posterior distributions) for the parameter values in Table 2.

5. Model Comparison to Photometric Variations

Time-series photometry of Spica in the literature is limited. Because the orbital period, P=4.015 days, is nearly an integer number of days, the space-based Hipparcos photometric data, obtained in the wide H_p-band (van Leeuwen et al. 1997), provide the best orbital phase coverage with 72 measurements from 1990 and 1993. Ground-based differential photometry of Spica relative to θ Vir is available in the V-band (494 measurements) from Shobbrook et al. (1969). These data shows the best comparison between the 4-hour β-Cephei variation and the 4-day ellipsoidal variation. The Strömgren *uvby* data (13 measurements) from Sterken et al. (1986) provide another check on the amplitude of the ellipsoidal variation.

The Hipparcos and V-band data are compared to our best fit model predictions in Figure 8. The shorter period β-Cephei variations and longer period ellipsoidal variations are clear from the V-band data and help explain the scatter in the Hipparcos light curve. Our model doesn't not include pulsational variability. This comparison to the ellipsoidal variation shows that the predicted variation in the projected size of the primary star is generally consistent with the photometry. Figure 9 shows the best-fit model appears to be consistent with the y-band variations, within the scatter caused by the pulsational variability.

6. Discussion

6.1. Apsidal Constant

For Spica, the advance of periastron is most sensitive to the mass distribution within the primary because the contribution to the apsidal motion is proportional to the the fifth power of the ratio of equatorial radius to the semi-major axis. The observational apsidal constant is computed from

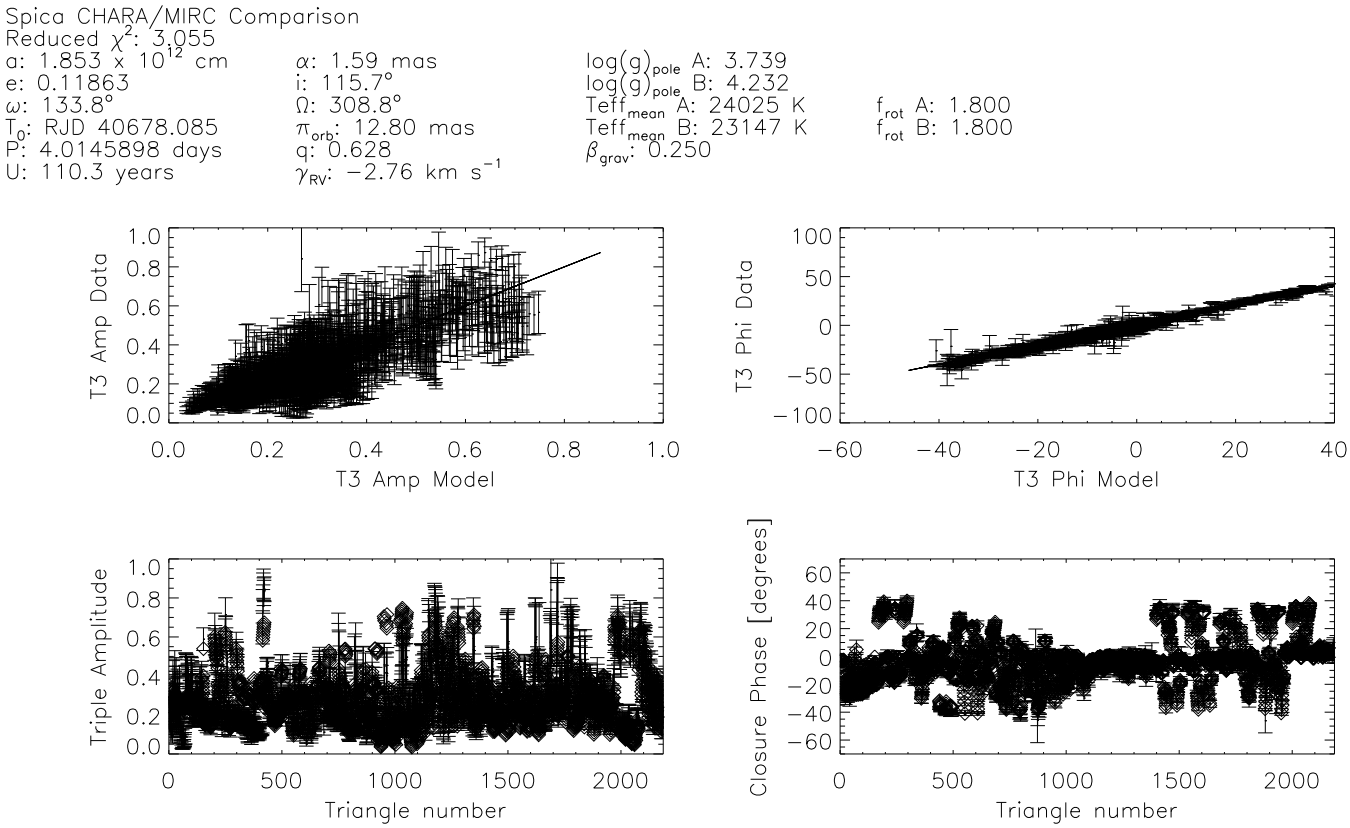

Fig. 2.— MIRC triple amplitude (T3 Amp) and closure phase (T3 Phi) data in comparison to the global best fit.

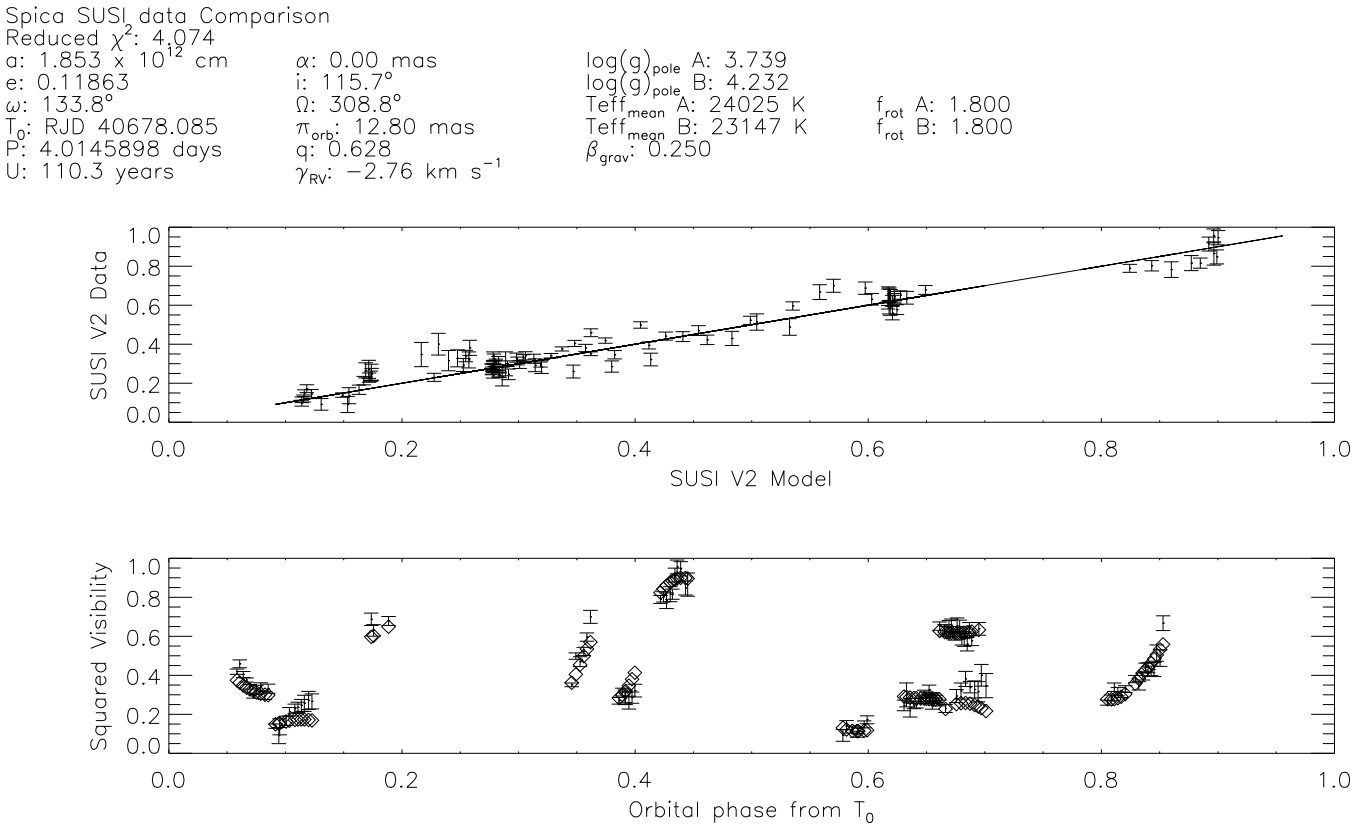

Fig. 3.— SUSI squared visibility data in comparison to the global best fit.

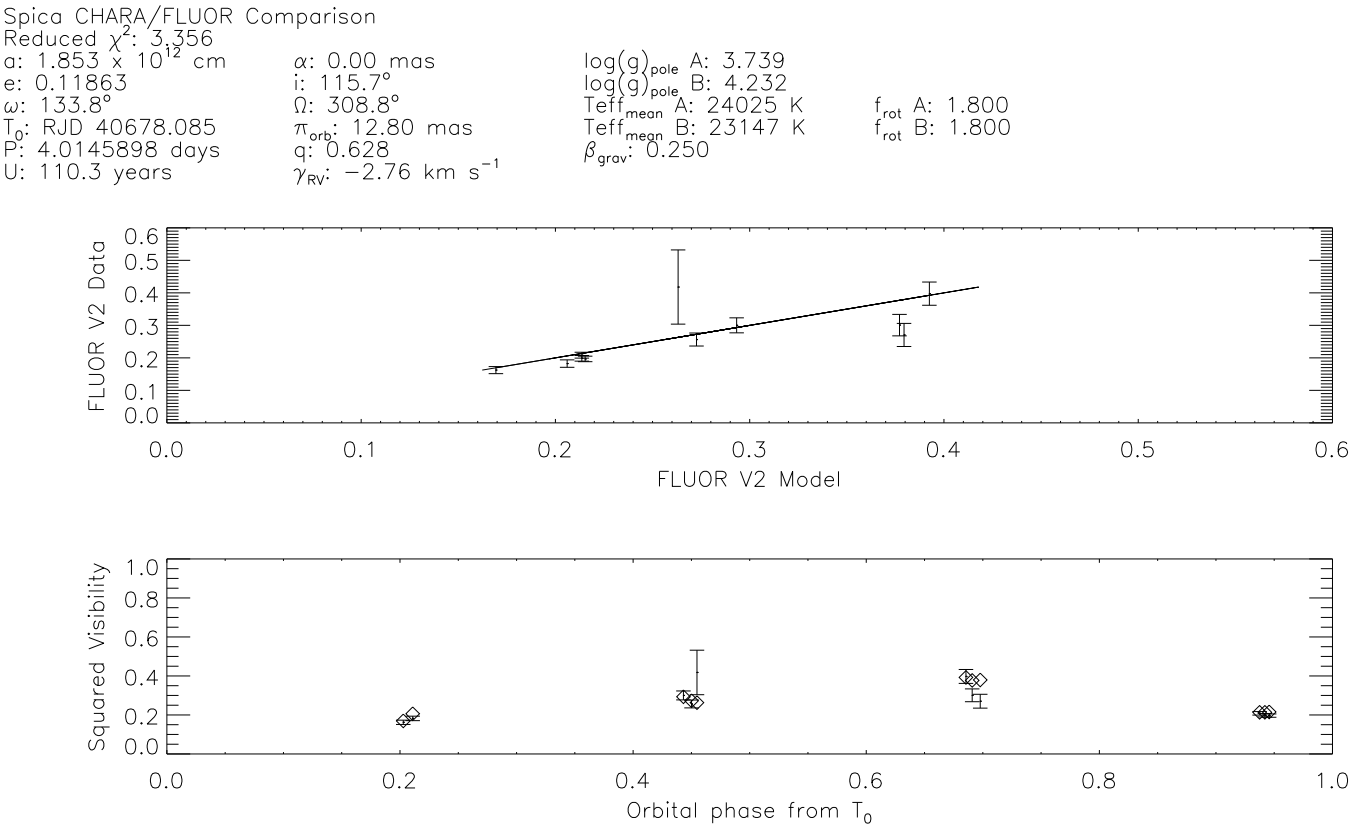

Fig. 4.— CHARA/FLUOR squared visibility data in comparison to a model with the global best fit parameters.

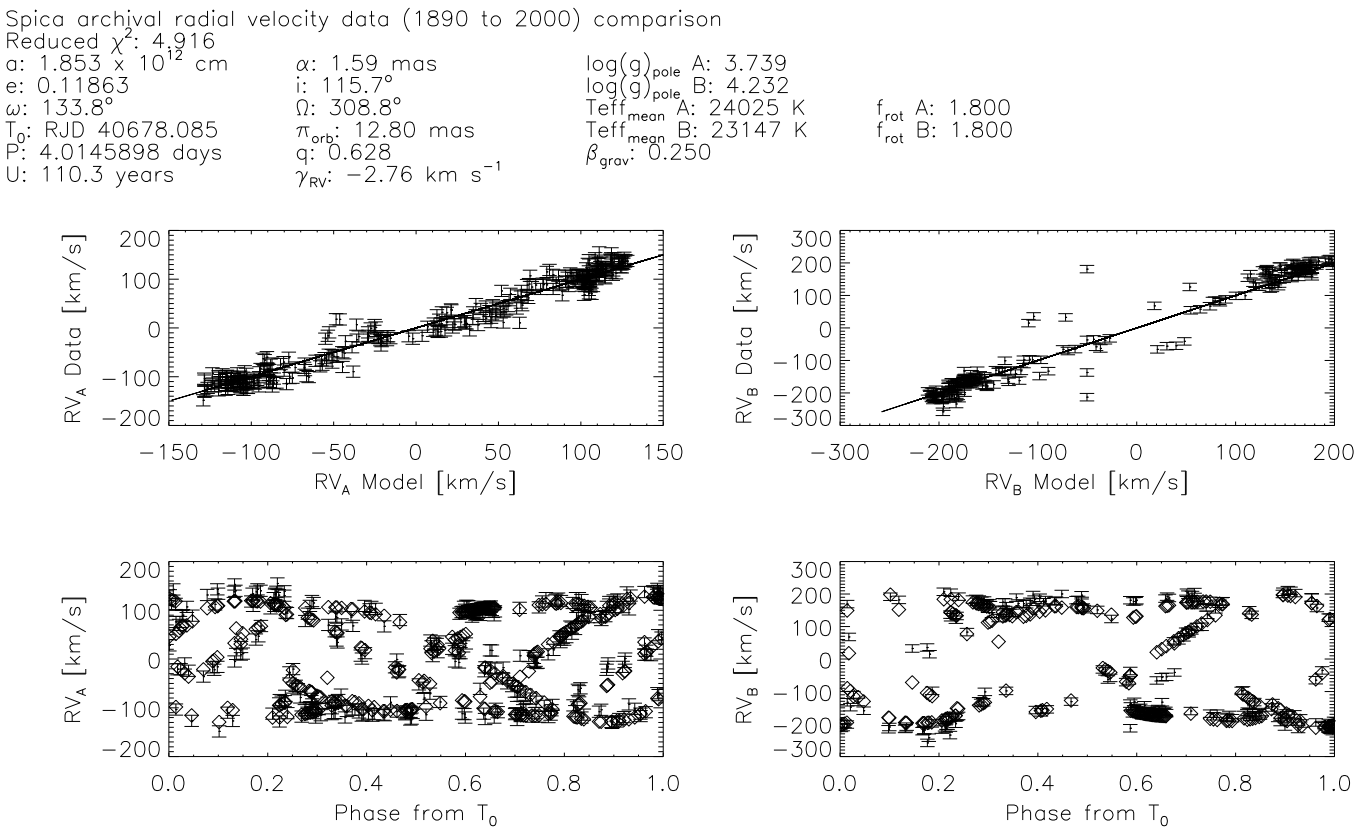

Fig. 5.— A comparison to 577 radial velocity measurements (340 for Spica A, 237 for Spica B) from Vogel (1890); Baker (1910); Struve & Ebbighausen (1934); Struve et al. (1958); Shobbrook et al. (1972); Riddle (2000) in comparison with the model with the global best fit parameters. Due to apsidal motion, data from significantly different epochs will be shifted relative to one another. High residuals are expected for Spica B at velocities below 100 km s^{-}1 due to blending with the much stronger lines of Spica A.

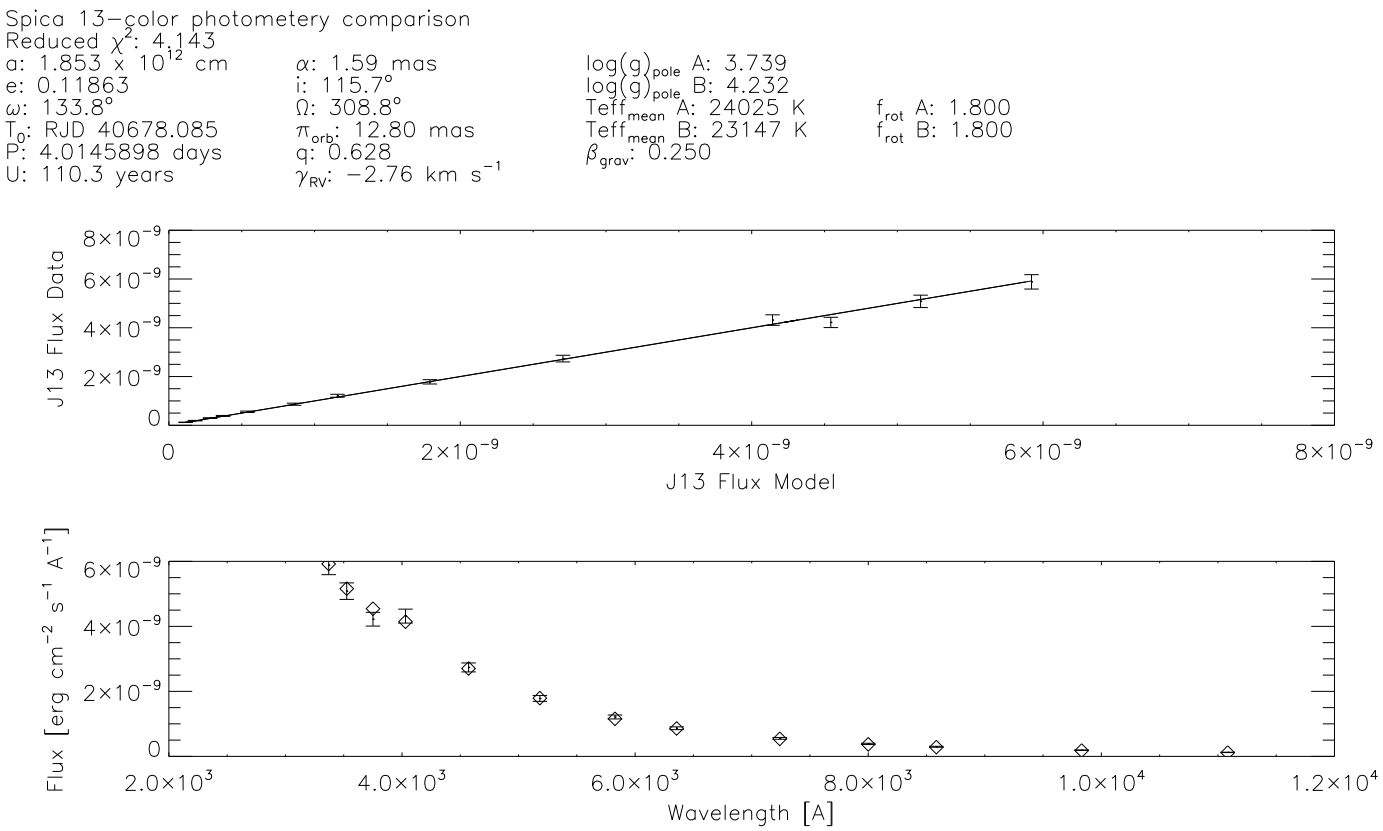

Fig. 6.— Johnson 13-color photometry (Johnson & Mitchell 1975) in comparison to a model with the global best fit parameters.

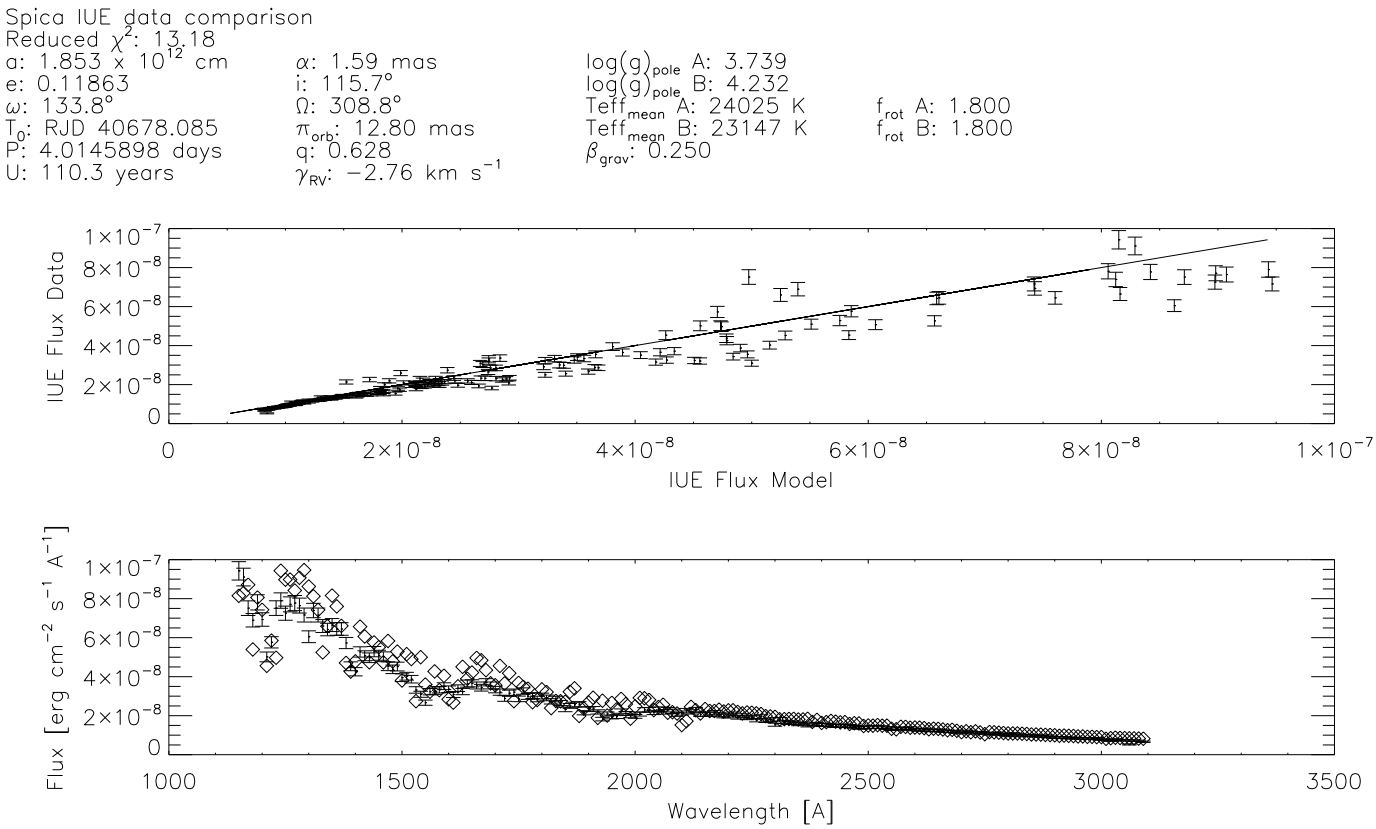

Fig. 7.— IUE rebinned high-resolution data (SWP33082RL and LWP12841RL, see e.g. González-Riestra et al. (2000))in comparion to a model with the global best fit parameters.

Fig. 8.— The top panel shows a comparison between the synthetic H_p magnitudes (diamonds) from the best-fit model and the Hipparocos photometry (van Leeuwen et al. 1997) shown as error bars. The lower panel shows the differential variation between θ Vir, assumed to be of constant magnitude, and Spica from Shobbrook et al. (1969) in comparison to synthetic V-band photometry from our best-fit model computed as 50 points over one period at the mean epoch of the observations. A comparison of the top and bottom panels shows the advancement of periastron between 1968 and 1990s.

Fig. 9.— A comparison between synthetic Strömgren y magnitudes (diamonds) from the best-fit model, computed as 50 points over one period at the mean epoch of the observations, and the Sterken et al. (1986) differential y photometry relative to θ Vir.

$$\bar{k_2} = \frac{1}{c_{12}+c_{22}}\frac{P}{U} \quad (1)$$

$$c_{21} = (((\frac{\omega_{1\ \mathrm{rot}}}{\omega_{\mathrm{orb}}})^2(1+q)f(e)) + 15qg(e))(\frac{R_{1\ \mathrm{equ}}}{a})^5 \quad (2)$$

$$c_{22} = (((\frac{\omega_{2\ \mathrm{rot}}}{\omega_{\mathrm{orb}}})^2(1+1/q)f(e)) + 15(1/q)g(e))(\frac{R_{2\ \mathrm{equ}}}{a})^5 \quad (3)$$

$$f(e) = (1-e^2)^{-2} \quad (4)$$

$$g(e) = \frac{f(e)^{5/2}}{8}(8+(12e^2)+(e^4)) \quad (5)$$

where P is the orbital period, U is the apsidal period, $\omega_{1\ \mathrm{rot}}$ and $\omega_{1\ \mathrm{rot}}$ are the angular rotation frequencies of the primary and secondary, $\omega_{\mathrm{orb}} = 2\pi/P$ is the angular frequency of the orbit, $q = M_2/M_1$ is the mass ratio, $R_{1\ \mathrm{equ}}$ and $R_{1\ \mathrm{equ}}$ are the equatorial radii of the primary and secondary, and e is the orbital eccentricity.

The mean equatorial radius of the primary (averaging the point, the two side, and the back radii) varies from 7.93 $R_\odot$(periastron) to 8.01 $R_\odot$(apastron). The primary polar radius is fixed at 7.40 $R_\odot$. The mean equatorial radius of the secondary, 3.76 $R_\odot$, varies by less than 1% throughout the orbit. The apsidal constant $\bar{k_2}$ is dominated by c_{21} term by a factor of approximately 40-to-1. The best fit parameters yield $\log(\bar{k_2}) = -2.50$ which is a larger value than range -2.8 to -2.6 adopted by Mathis & Odell (1973).

The parameter $\bar{k_2}$ takes the value 0.75 for homogeneous stars and zero for point masses. Therefore, $\log(\bar{k_2})$ takes values from -0.125 to -∞. As a star evolves, the distribution of mass becomes more centrally concentrated and $\log(\bar{k_2})$ becomes increasingly negative. Batten (1973) shows that for a star similar to Spica A, it should start on the ZAMS at $\log(\bar{k_2}) = -1.9$. Once the core hydrogen is reduced from 60% to 10% this value becomes -2.25. The value of $\log(\bar{k_2})$ from our orbital solution yields a still smaller value indicating the star is more centrally concentrated than expected. Claret shows the expected value for $\bar{k_2}$ from evolution theory should be... [insert value here].

The lower eccentricity raises $\log(\bar{k_2})$ by 0.05 dex, while the higher asynchronous rotation factors lower $\log(\bar{k_2})$ by [xxx] dex.

7. Conclusions

A. Model Construction

To analyze new interferometric data together with archival radial velocity and spectrophometric, one of us (JPA) has developed detailed model atmospheres for the components including tidal distortion, rotational distortion, and the resulting gravity darkening. Our code, written in the Interactive Data Language (IDL), follows the code of Wilson (1979) and computes the surfaces of binary stars in close, eccentric orbits from an approximate gravitational potential and interpolates model atmosphere intensities onto these surfaces. The surfaces are then projected onto the plane of the sky to make a synthetic intensity map from which synthetic interferometric observables (squared visibilities, closure phases, triple amplitudes) can be computed. The code also predicts the photometric variability associated the changing projected area of the primary star (ellipsoidal variation).

A.1. Input Parameters

A synthetic image of Spica's components is constructed from 16 input parameters. The range of parameter space explored for a best-fit solution is limited to values from the literature as discussed parameter below.

P orbital period taken from periastron to periastron. The value is fixed at 4.01454 days from the Shobbrook et al. (1972, see Table IV) value 4.01454 ± 0.00003 days.

T_0 epoch of periastron. The value JD 2440678.09 is adopted for comparison with (Herbison-Evans et al. 1971, see Table III)

q ratio of the secondary mass to the primary mass. Constrained by the ratio of the semi-amplitudes K_1 and K_2 from the radial velocity curves. Shobbrook et al. (1972) establish the ratio 0.629 ± 0.059, confirmed by Riddle (2000) who finds 0.6359 ± 0.0381. The range of q is limited to between 0.52 and 0.75.

e eccentricity. Spectroscopic limits on e from Struve et al. (1958) range from 0.097 to 0.172. The values were narrowed to 0.146 ± 0.03 by Shobbrook et al. (1972) and later to 0.067 ± 0.014 by Riddle (2000). e is limited to between 0.0 and 0.18.

ω longitude of periastron. Herbison-Evans et al. (1971) adopted $\omega = 138°$ at T_0. Based on values published by Shobbrook et al. (1972), the uncertainty in this value is approximately $\pm 10°$. ω is limited to between 128°and 148°.

U apsidal period. Figure 10 shows a linear fit to the evolution of ω over approximately 50 years. This fit yields an estimate for the apsidal period of $U = 139 \pm 7$ years, the period for ω is advance through 360°. Shobbrook et al. (1972) adopted $U = 128 \pm 12$ years. U is limited to between 116 and 148 years.

Ω the position angle of the line of nodes. Established first by Herbison-Evans et al. (1971) to be $131\overset{\circ}{.}6 \pm 2\overset{\circ}{.}1$. This value is constrained by interferometry alone and is therefore model dependent. Ω is limited to between 118°and 148°.

i orbital inclination. Like Ω, this parameter was constrained to $65\overset{\circ}{.}9 \pm 1\overset{\circ}{.}8$ by Herbison-Evans et al. (1971) Our simulations indicate the values $i \geq 68°$ lead to eclipses not observed thus establishing a rough upperlimit. i limited to between 45°and 68°. For clockwise motion on the sky this corresponds to 112°to 135°.

d distance to Spica. The original Hipparcos parallax of Spica, $\pi_{\rm hipp} = 12.44 \pm 0.86$ mas (Perryman et al. 1997), based on a single star fit to the astrometric measurements, is consistent with the orbital parallax of Herbison-Evans et al. (1971), 11.90 ± 0.56 mas. The new Hipparcos reduction (van Leeuwen 2007) moves Spica a bit closer with a parallax of $\pi_{\rm hipp} = 13.06 \pm 0.70$ mas, less consistant with the orbital parallax from the Herbison-Evans et al. (1971) solution.

a semi-major axis. $a \sin i$ is constrained by spectroscopy to $1.76 \pm 0.08 \times 10^{12}$ cm by Shobbrook et al. (1972) and confirmed by Riddle (2000) $1.730 \pm 0.058 \times 10^{12}$ cm. a is limited to between 1.5×10^{12} cm to 2.2×10^{12} cm to account for a range of inclinations.

$g_{1\ \rm pole}$ surface gravity at the pole of the primary. Spectroscopic determinations of the surface gravity generally assume the star is spherical, thus the surface gravity takes a single value. Lyubimkov et al. (1995) find $\log g = 3.7 \pm 0.1$ for the primary star. The surface gravity will be lower at the equator and higher at the pole. $\log g_{1\ \rm pole}$ is limited to between 3.67 and 3.95.

$g_{2\ \rm pole}$ surface gravity at the pole of the secondary. Lyubimkov et al. (1995) establish $\log g = 4.2 \pm 0.2$ for the secondary star. $\log g_{2\ \rm pole}$ is limited to between 3.90 and 4.25.

$T_{\rm eff\ 1\ pole}$ effective temperature at the pole of the primary. Lyubimkov et al. (1995) establish $T_{\rm eff} = 24700 \pm 500$ K for the primary star. The $T_{\rm eff\ 1\ pole}$ is limited to between 22525 K and 25550 K.

$T_{\rm eff\ 2\ pole}$ effective temperature at the pole of the secondary. Lyubimkov et al. (1995) establish $T_{\rm eff} = 20800 \pm 1500$ K for the secondary star. $T_{\rm eff\ 2\ pole}$ is limited to between 18750 K and 24450 K.

$\omega_{1\ \rm rot}$ angular rotation rate of the primary. This quantity, equal to the ratio of the mean equatorial rotation velocity to the mean equatorial radius, is constrained by $v \sin i$ and radius estimates. Smith (1985) establishes $v \sin i = 161 \pm 2$ km s^{-1} based on high-dispersion profile analysis. Lyubimkov et al. (1995)

find a range of primary radii from 6.6 $R_\odot$ to 9.5 $R_\odot$. Herbison-Evans et al. (1971) find $R_1 = 8.1 \pm 0.5$ $R_\odot$. With these values and a range inclinations from 45 °to 68°yield a range for asynchronous rotation rates, $\frac{\omega_{1\ \mathrm{rot}}}{\omega_{orb}}$, from 1.5 to 2.2 [check this!]

$\omega_{2\ \mathrm{rot}}$ angular rotation rate of the secondary. Smith (1985) establishes $v \sin i = 70 \pm 5$ km s^{-1} for the secondary and estimates 4.4± 0.7 $R_\odot$ for the radius of the secondary. These values lead to $\frac{\omega_{1\ \mathrm{rot}}}{\omega_{orb}}$ from 1.2 to 1.7 [check this].

The angular rates $\omega_{1\ \mathrm{rot}}$ and $\omega_{2\ \mathrm{rot}}$ rates are tied to the inclination as constrained by the observed $v \sin i$ values and to the equatorial radius which depends on the input gravities and the potential field which in turn depends on these angular rotation rates.

All parameters with the exception of P and T_0 have been adjusted to best match a simultaneous fit to the interferometric (SUSI,CHARA/FLUOR,CHARA/MICRC) and spectrophotometric (13-color photometry) (Johnson & Mitchell 1975) data sets. The iteration of these input parameters in search of the lowest reduced chi-square, χ^2_ν, is performed with the Interactive Data Language (IDL) routine `MPFITFUN`[1] which employs a Levenberg-Marquard minimization technique. The IDL routine `TRIAXIAL`, for triaxial stars in tidally distorted binary systems, is called by `MPFITFUN` and takes the input parameters listed above and generates synthetic interferometric and spectrophotometric data including high-resolution ($R \sim 100{,}000$) spectra.

The initial starting values were chosen from the Herbison-Evans et al. (1971) solution and limits on the parameters adopted as discussed above.

A.2. The Long Road to Convergence....

Initially the value for a was limited by the Herbison-Evans et al. (1971) values $1.87 - 1.99 \times 10^{12}$ cm. During the fit the value for the upperlimit was increased to 2.5×10^{12} cm when this parameter became pegged at the original upper limit, through it settled down to 1.92×10^{12} cm.

A.2.1. Failure of Levenberg-Marquard...

A.2.2. Genetic Algorithms...

A.3. Coordinate System and Potentials

The `TRIAXIAL` model is an improvement over the standard interferometric model of two uniform disks. The code predicts the interferometric observables from a Fourier transform of a synthetic intensity map of the Spica system. The construction of synthetic light curves binaries with eccentric orbits and asynchronous rotation was pioneered by Wilson (1979). For photometric work, the components need not be modeled as spatially resolved. For the interferometric predictions, triaxial figures of the components must be projected into the plane of the sky in order to compute the Fourier transform. For circular orbits with synchronous rotating components, the surface forms of the stellar components are defined by equipotential surfaces of centrally-condensed point masses (Roche Models). For the case of eccentric orbits like Spica, the distance between the centers components changes with orbital phase, so the equipotential surfaces must be computed at each phase point under the assumption that each component readjusts to equilibrium on time scales must faster than the orbital period. Adding uniformly rotating yet asynchronous stellar components has no consistent simple solution. To established constant-density surfaces for the components, we start with the

[1]This routine from C.B. Markwardt is available at http://cow.physics.wisc.edu/ craigm/idl/idl.html

Fig. 10.— A linear fit to evolution of Spica's longitude of periastron with time. ω values from Luyten & Ebbighausen (1935), Struve et al. (1958), Dukes (1974), Lyubimkov et al. (1995), and Riddle (2000). The best-fit apsidal period, U, is 130±8 years with a reduced $\chi^2 = 2.8$.

approximate potential from (Limber 1963, see equation 16) understanding that it is not strictly valid due to the neglect of small Coriolis forces from stellar fluid flow. This potential assumes that both rotational angular velocity vectors and the orbital angular velocity vector are parallel.

The potential describing the surface of the primary in rectangular coordinates is

$$V_1 = -G\frac{m_1}{r_1} - G\frac{m_2}{r_2} - \frac{\omega_{\rm orb}^2}{2}\left[\left(\frac{\omega_{1\ \rm rot}}{\omega_{\rm orb}}\right)^2 - 1\right](x^2+y^2) - \frac{1}{2}\omega_{\rm orb}^2\left[\left(x - \frac{Dm_2}{m_1+m_2}\right)^2 + y^2\right] \tag{A1}$$

where the origin of the coordinate system is at the center of primary, in the frame which rotates about an axis perpendicular to the orbital plane at a constant rate with the period of the binary, P. The x-axis is in the orbital plane along the line of centers. The first term in equation A1 represents the potential from the primary where G is the gravitational constant, m_1 the mass of the primary, and

$$r_1 = (x^2+y^2+z^2)^{1/2} \tag{A2}$$

is the distance from the center of mass of the primary. The second term represents the potential from the secondary mass m_2 where

$$r_2 = \left((D-x)^2+y^2+z^2\right)^{1/2} \tag{A3}$$

is the distance from the center of mass of the secondary, D is the instantaneous separation of the centers of mass of the two stars. In units of the semi-major axis, D is computed from the eccentricity e and the true anomaly υ:

$$D = \frac{1-e^2}{1+e\cos\upsilon}. \tag{A4}$$

The primary point mass (x, y, z) coordinate is at (0,0,0) and the secondary is at $(D, 0, 0)$. The third term in equation A1 represents the contribution from the asynchronous rotation where $\omega_{\rm orb}$ is the angular orbital rate $(2\pi/P)$ and $\omega_{1\ \rm rot}$ is the angular rotational rate of the primary. This term vanishes for syncronicity: $\omega_{1\ \rm rot} = \omega_{\rm orb}$. The forth term represents potential from the centrifugal acceleration due to the rotating frame.

For this asynchronous system a separate potential is needed to describe the surface of the secondary in rectangular coordinates:

$$V_2 = -G\frac{m_1}{r_1} - G\frac{m_2}{r_2} - \frac{\omega_{\rm orb}^2}{2}\left[\left(\frac{\omega_{2\ \rm rot}}{\omega_{\rm orb}}\right)^2 - 1)\right]\left((D-x)^2+y^2\right) - \frac{1}{2}\omega_{\rm orb}^2\left[\left(x - \frac{Dm_2}{m_1+m_2}\right)^2 + y^2\right] \tag{A5}$$

Both potentials (equations A1 and A5) can be greatly simplified when put in terms of the mass ratio $q = m_2/m_1$, instantaneous separation D, the respective radius vectors r_1 and r_2, the direction cosines of radius vector,

$$\lambda = \cos\varphi\sin\vartheta \tag{A6}$$
$$\mu = \sin\varphi\sin\vartheta \tag{A7}$$
$$\nu = \cos\vartheta \tag{A8}$$

where φ is the azimuthal angle (longitude) and ϑ is the polar angle (co-latitude). The rectangular coordinates are related to the spherical coordinates by

$$x = r_1\ \lambda \tag{A9}$$
$$x = D + r_2\ \lambda \tag{A10}$$
$$y = r_1\ \mu \tag{A11}$$
$$y = r_2\ \mu \tag{A12}$$
$$z = r_1\ \nu \tag{A13}$$
$$z = r_2\ \nu. \tag{A14}$$

The simplified potentials,

$$\Omega_1 = \frac{-V_1}{Gm_1} - \frac{1}{2D}\left(\frac{q^2}{1+q}\right) \tag{A15}$$

$$\Omega_2 = \frac{-V_2}{Gm_1} - \frac{1}{2D}\left(\frac{q^2}{1+q}\right) \tag{A16}$$

for the primary and secondary stars respectively, differ from the original potentials V_1 and V_2 by constants. The mass ratio q is fixed for the model and the instantaneous separation D (equation A4) is fixed at each time step. With the semi-major axis set to unity, Kepler's third law becomes,

$$\omega_{\rm orb}^2 = Gm_1(1+q), \tag{A17}$$

and the simplified potentials in terms of q, D, r_1, r_2, and the direction cosines are

$$\Omega_1 = \frac{1}{r_1} + \frac{q}{\sqrt{D^2 + r_1^2 - 2r_1\lambda D}} - qr_1\lambda D + \frac{1}{2}\left(\frac{\omega_{1\ \rm rot}}{\omega_{\rm orb}}\right)^2 (1+q)r_1^2(1-\nu^2) \tag{A18}$$

$$\Omega_2 = \frac{1}{\sqrt{D^2 + r_2^2 + 2r_2\lambda D}} + \frac{q}{r_2} + \frac{1}{2}(1+q)(D^2 + 2Dr_2\lambda) - q(D^2 + r_2\lambda D) + \frac{1}{2}\left(\frac{\omega_{2\ \rm rot}}{\omega_{\rm orb}}\right)^2 (1+q)r_2^2(1-\nu^2) \tag{A19}$$

where all lengths are in units of the semi-major axis a.

A.4. Solving for the Stellar Surfaces

The computation of the surface form for each star begins at the pole, starting with the polar radius

$$r_{1\ \rm pole} = \sqrt{\frac{Gm_1}{g_{1\ \rm pole}}} \tag{A20}$$

$$r_{2\ \rm pole} = \sqrt{\frac{Gm_2}{g_{2\ \rm pole}}} \tag{A21}$$

for the primary and secondary respectively, where $\log(g_{1\ \rm pole})$ and $\log(g_{2\ \rm pole})$ are the polar surface gravities of the components.The mass of the primary is obtained from Kepler's third law,

$$m_1 = \frac{4\pi^2 a^3}{GP^2(1+q)} \tag{A22}$$

$$m_2 = m_1 q \tag{A23}$$

The values of the simplified potentials at the pole ($\vartheta = 0$) of each star from equations A18 and A19 are

$$\Omega_{1\ \rm surface} = \frac{1}{r_{1\ \rm pole}} + \frac{q}{\sqrt{D^2 + r_{1\ \rm pole}^2}} \tag{A24}$$

$$\Omega_{2\ \rm surface} = \frac{1}{\sqrt{D^2 + r_{2\ \rm pole}^2}} + \frac{q}{r_{2\ \rm pole}} + \frac{1}{2}(1+q)D^2 - qD^2 \tag{A25}$$

A Newton-Raphson iteration is used to find the radius at the next co-latitude starting from the pole:

$$r_{\rm new} = r_{\rm guess} - \frac{\left(\Omega(r_{\rm guess}, \varphi, \vartheta) - \Omega_{\rm surface}\right)}{\partial\Omega/\partial r} \tag{A26}$$

where $\Omega(r_{\rm guess}, \phi, \theta)$ is the value computed from equation A18 in the case of the primary or A19 in the case of the secondary and $\Omega_{\rm surface}$ corresponds to the constant (for each time step) from A24 or A25. The value of $r_{\rm new}$ becomes $r_{\rm guess}$ for the next iteration until the difference $r_{\rm new} - r_{\rm guess}$ is less than 10^{-6}. In practice the iteration always converges to a difference just less than 10^{-4}.

The simplified potential derivatives with respect to the radius vector are

$$\frac{\partial \Omega_1}{\partial r_1} = -\frac{1}{r_1^2} - \frac{q(r_1 - \lambda D)}{(D^2 + r_1^2 - 2r_1\lambda D)^{3/2}} - q\lambda D + \left(\frac{\omega_{1\ \rm rot}}{\omega_{\rm orb}}\right)^2 (1+q) r_1 (1-\nu^2) \tag{A27}$$

$$\frac{\partial \Omega_2}{\partial r_2} = -\frac{D\lambda + r_2}{(D^2 + r_2^2 + 2r_2\lambda D)^{3/2}} - \frac{q}{r_2^2} + \lambda D + \left(\frac{\omega_{2\ \rm rot}}{\omega_{\rm orb}}\right)^2 (1+q) r_2 (1-\nu^2). \tag{A28}$$

$$\tag{A29}$$

The surface gravity at each point on the surface is most easily computed from components in rectangular coordinates,

$$g_x = \frac{\partial \Omega_1}{\partial x} = \frac{-x}{(x^2+y^2+z^2)^{3/2}} + \frac{q(D-x)}{((D-x)^2+y^2+z^2)^{3/2}} + \left(\frac{\omega_{1\ \rm rot}}{\omega_{\rm orb}}\right)^2 (1+q)x - qD \tag{A30}$$

$$g_x = \frac{\partial \Omega_2}{\partial x} = \frac{-x}{(x^2+y^2+z^2)^{3/2}} + \frac{q(D-x)}{((D-x)^2+y^2+z^2)^{3/2}} + \left[1 - \left(\frac{\omega_{2\ \rm rot}}{\omega_{\rm orb}}\right)^2\right](1+q)(D-x) + x(1+q) - qD \tag{A31}$$

$$g_y = \frac{\partial \Omega_{1,2}}{\partial y} = \frac{-y}{(x^2+y^2+z^2)^{3/2}} + \frac{-yq}{((D-x)^2+y^2+z^2)^{3/2}} + \left(\frac{\omega_{1,2\ \rm rot}}{\omega_{\rm orb}}\right)^2 (1+q)y \tag{A32}$$

$$g_z = \frac{\partial \Omega}{\partial z} = \frac{-z}{(x^2+y^2+z^2)^{3/2}} + \frac{-zq}{((D-x)^2+y^2+z^2)^{3/2}} \tag{A33}$$

where the effective surface gravity then is

$$g_{\rm eff} = (g_x^2 + g_y^2 + g_z^2)^{1/2}. \tag{A34}$$

Following Von Zeipel's purely radiative law, the effective temperature at each point on the surface is computed from

$$T_{\rm eff}(r_{1,2}, \phi, \theta) = T_{\rm eff\ 1,2\ pole} \left(\frac{g_{\rm eff}(r_{1,2}, \phi, \theta)}{g_{1,2\ \rm pole}}\right)^{1/4}. \tag{A35}$$

Following Kopal (1959, equation 2-6, page 175), the direction cosines of the normal to surface at each point, which differ from the direction cosines of the radius vector for non-spherical stars, are

$$l = g_x / g_{\rm eff} \tag{A36}$$

$$m = g_y / g_{\rm eff} \tag{A37}$$

$$n = g_z / g_{\rm eff} \tag{A38}$$

¿From (Kopal 1959, equation 2-3, page 174), the cosine of the angle between the radius vector and the surface normal is

$$\cos\beta = \lambda l + \mu m + \nu n \tag{A39}$$

which is used below to compute the flux and the cosine for the emergent radiation for the observer at time t relative to the surface normal is

$$\mu(\varphi, \vartheta, t) = l l_0 + m m_0 + n n_0 \tag{A40}$$

where l_0, m_0, n_0 denote the direction cosines for the line of sight

$$l_0 = -\sin\psi \sin i \tag{A41}$$

$$m_0 = \cos\psi \sin i \tag{A42}$$

$$n_0 = \cos i. \tag{A43}$$

These direction cosine formulae differ from (Kopal 1959, equation 2-5, page 174) where ψ is measured from the moment of superior conjunction, when the primary eclipses the secondary for high inclination systems. We define angle ψ as

$$\psi = \omega + \upsilon \tag{A44}$$

the position of the secondary star from the line of nodes, which is less than Kopal (1959) by 90 degrees. The angle ω, the longitude of periastron, for each time step t is

$$\omega = \omega_0 + \left(\frac{360}{365.25}\right)\frac{t - T_0}{U} \tag{A45}$$

where ω_0 is the longitude of periastron at a reference epoch T_0 and U is the apsidal period.

The true anomaly is calculated following Herbison-Evans et al. (1971),

$$\tan\frac{\upsilon}{2} = \Big[(1+e)/(1-e)\Big]^{1/2}\tan\frac{E}{2} \tag{A46}$$

where the eccentric anomaly E from the first order approximation is

$$E = M + \frac{e\sin M}{1 - e\cos M} \tag{A47}$$

and M is the mean anomaly,

$$M = 2\pi\frac{t - T_0}{P}. \tag{A48}$$

A.5. Interpolation of the Rest Frame Intensities

Following Aufdenberg et al. (2006), the intensity at each point (ϕ, θ) on each stellar surface is interpolated from synthetic radiation fields calculated from two grids of spherical, hydrostatic `PHOENIX` (version 15.04.00E) stellar atmosphere models (Hauschildt et al. 1999) spanning 19000 K to 25500 K in $T_{\rm eff}$ and 3.30 to 3.95 in $\log(g)$ for the primary

$$T_{1j} = 19000 + 250\cdot j \quad \mathrm{K} \qquad j = \{0, 1, \ldots, 26\} \tag{A49}$$

$$\log(g_{1l}) = 3.30 + 0.05\cdot ly \qquad l = \{0, 1, \ldots, 13\}. \tag{A50}$$

(378 separate atmosphere models) and for the secondary

$$T_{2j} = 20000 + 250\cdot j \quad \mathrm{K} \qquad j = \{0, 1, \ldots, 13\} \tag{A51}$$

$$\log(g_{2l}) = 4.15 + 0.05\cdot l \qquad l = \{0, 1, \ldots, 3\}. \tag{A52}$$

(39 separate atmosphere models), both grids with a resolution of 250 K in $T_{\rm eff}$ and 0.05 in $\log(g)$.

At each rest wavelength, $I(\lambda, \mu)$ is evaluated at 64 angles by `PHOENIX`. The logarithm of the intensity I_λ from point on a star corresponding to a local effective temperature $T_{\rm eff}$ and effective surface gravity $\log(g)$ from corresponding emergent μ along the line of sight is obtained using the trivariate interpolation IDL routine `INTERPOLATE`.

A.6. Mapping Intensities on to the Plane of the Sky

An intensity $I_\lambda(\varphi, \vartheta)$ originates from a location (x, y, z) computed from equations A6-A14. Following Kopal (1959, see equations 3-13 to 3-16, pages 44-45) the coordinates of this location in the plane of the sky are

$$\begin{aligned} x_{\rm sky} &= x\big[-\cos(\omega+\upsilon)\cos\Omega + \cos i\sin\Omega\sin(\omega+\upsilon)\big] \\ &\quad + y\big[-\sin(\omega+\upsilon)\cos\Omega - \cos i\sin\Omega\cos(\omega+\upsilon)\big] + z\big[\sin i\sin\Omega\big] \end{aligned} \tag{A53}$$

$$\begin{aligned} y_{\rm sky} &= x\big[\cos(\omega+\upsilon)\sin\Omega + \cos i\cos\Omega\sin(\omega+\upsilon)\big] \\ &\quad + y\big[\sin(\omega+\upsilon)\sin\Omega + \cos i\cos\Omega\cos(\omega+\upsilon)\big] + z\big[\sin i\cos\Omega\big] \end{aligned} \tag{A54}$$

where the unsubscripted Ω is the orbital element, the position angle of the line of nodes, not to be confused with the potentials above. This coordinate transformation converts the x, y, and z locations on each star in the rotating frame, with the x-axis along a line towards the secondary and the z-axis perpendicular to the orbital plane, to the plane of the sky where the x-axis points north and the z-axis points to the observer. One additional rotation, 90°counter clockwise, brings the x-axis in line with the east.

After the transformation and rotation, $x_{\rm sky}$ and $y_{\rm sky}$ correspond to offsets in right ascension and declination ($\Delta\alpha$,$\Delta\delta$) relative to the the center of mass of the primary star in units of milliarcseconds. This mapping produces a synthetic image of each star at each time step t. Next, the intensity map of each star is regridded (using the IDL routine `TRIGRID`) from the original sampling based on the φ, ϑ grid onto a regular 512×512 grid of points in $x_{\rm sky}$ and $y_{\rm sky}$. Next, the two images are added to form a synthetic image of the binary.

A.7. Synthetic Visibility and Closure Phase Compuation

We evaluate a set of discrete 2-D Fourier transforms to compute synthetic squared visibilities for comparison with the CHARA/FLUOR and SUSI data sets and synthetic closure phases for comparison with the CHARA/MIRC data sets.

The first step is to compute the real (cosine) and imaginary (sine) components of the discrete Fourier transform for each wavelength at each of the spatial frequency coordinates (u, v) corresponding to the projected baseline and orientation of each data point. The (u, v) coordinates of each telescope pair, in units of meters, are converted to the corresponding spatial frequency coordinates (u_k, v_k) in units of cycles per arcsecond for each wavelength λ_k. The Fourier transform,

$$V_\lambda^2(u,v) = \left[\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} S_\lambda I_\lambda(x_{\rm sky}, y_{\rm sky}) e^{-i2\pi(u\,x_{\rm sky}+v\,y_{\rm sky})}\,{\rm d}x_{\rm sky}\,{\rm d}y_{\rm sky}\right]^2 \tag{A55}$$

is approximated by the integration rule of Gaussian quadrature (e.g., Stroud & Secrest 1966; Press et al. 1992)

$$\begin{aligned} V_k^2(u_k, v_k) \quad \approx \quad & \left[\sum_{i=1}^{N} A_i \sum_{j=1}^{N} A_j I_k(x_i, y_j)\cos(2\pi(u_k x_i + v_k y_j))\right]^2 \\ + \quad & \left[\sum_{i=1}^{N} A_i \sum_{j=1}^{N} A_j I_k(x_i, y_j)\sin(2\pi(u_k x_i + v_k y_j))\right]^2 \end{aligned} \tag{A56}$$

where A_i, A_j and x_i, y_j are the weights and nodes of the quadrature, respectively. For our square grid, the $x-$ and $y-$coordinate nodes and weights are identical. The 2-D Gaussian quadrature is performed with a version of the IDL routine `INT_2D` modified to use an arbitrarily high number of nodes. The intensity at wavelength k, $I_k(x, y)$, is interpolated at (x_i, y_j) from the regular 512×512 spacing to the quadrature node points using the IDL routine `INTERPOLATE` which uses a cubic convolution interpolation method employing 16 neighboring points. The synthetic visibility is normalized to unity at zero spatial frequency by:

$$V_k(0,0) \approx \sum_{i=1}^{N} A_i \sum_{j=1}^{N} A_j I_k(x_i, y_j) \tag{A57}$$

We find $N = 512$ provides an accuracy of better than 1% for $V^2 > 10^{-3}$ (Aufdenberg et al. 2006).

For comparison with the CHARA/MIRC data, phase for each baseline

$$\phi(u_k, v_k) \approx \tan^{-1}\left[\frac{\sum_{i=1}^{N} A_i \sum_{j=1}^{N} A_j S_k I_k(x_i, y_j)\sin(2\pi(u_k x_i + v_k y_j))}{\sum_{i=1}^{N} A_i \sum_{j=1}^{N} A_j S_k I_k(x_i, y_j)\cos(2\pi(u_k x_i + v_k y_j))}\right] \tag{A58}$$

is computed using the ratio of the imaginary part to the real part of the discrete transform. The closure phase (Monnier 2000),

$$\Phi_{123} = \phi(u_{12}, v_{12})_k + \phi(u_{23}, v_{23})_k + \phi(u_{31}, v_{31})_k \tag{A59}$$

at each wavelength for each triangle of telescopes, labeled 1, 2 and 3, is computed as the sum of phases for three baselines that form the triangle. The corresponding triple amplitude is

$$V_{123} = \left(\left[V_k^2(u_{12}, v_{12})\right]^{1/2} \cdot \left[V_k^2(u_{23}, v_{23})\right]^{1/2} \cdot \left[V_k^2(u_{31}, v_{31})\right]^{1/2} \right) \left(V_k(0,0)\right)^{-3} \tag{A60}$$

where each V_k^2 is computed from equation A56 for each pair of telescopes that form the triangle.

A.8. Computation of the Observer's Frame Spectrum

The projected velocities of a surface element on each the two components are

$$v_1(r_1, \varphi, \vartheta) = -ar_1(\varphi, \vartheta)(\omega_{1\ \mathrm{rot}})(\lambda m_0 - \mu l_0) + K_1\big(e\cos\omega + \cos(\upsilon + \omega)\big) \tag{A61}$$

$$v_2(r_2, \varphi, \vartheta) = -ar_2(\varphi, \vartheta(\omega_{2\ \mathrm{rot}})(\lambda m_0 - \mu l_0) + K_2\big(e\cos(\omega + \pi) + \cos(\upsilon + \omega + \pi)\big) \tag{A62}$$

where the radii r_1 and r_2 are in units of the semi-major axis a. The first term follows Wilson & Sofia (1976, see equation 1) except we do not assume synchronization, so $\omega_{1,2\ \mathrm{rot}} \neq \omega_{\mathrm{orb}}$. The second term, due to the orbital motion of each component, is from Binnendijk (1960, equation 197, page 177) where the projected semi-amplitude velocities for the primary and secondary are

$$K_1 = \frac{2\pi}{P}\left(\frac{a}{1+q}\right)\frac{\sin i}{(1-e^2)^{1/2}} \tag{A63}$$

$$K_2 = \frac{2\pi}{P}\left(\frac{a}{1+(1/q)}\right)\frac{\sin i}{(1-e^2)^{1/2}} \tag{A64}$$

$$\tag{A65}$$

from Binnendijk (1960, equation 162, page 151). It is assumed that the rotational angular momentum vectors and orbital angular momentum vector are parallel such that the stars rotate in the same sense that they orbit the center of mass.

For each surface element, the observed wavelength is calculated

$$\lambda_{1,2\ \mathrm{obs}} = (1 + v_{1,2}/c)\lambda_{\mathrm{rest}} \tag{A66}$$

where c is the speed of light in vacuum and λ_{rest} is the vacuum rest-frame wavelength from model atmosphere grid. The intensity at a single observed wavelength is the sum of intensities from a range of rest-frame wavelengths which are Doppler shifted by rotational and orbital motion. An observer-frame wavelength grid is constructed by truncating the rest-frame wavelength grid at both the blue and red ends such that at every observer-frame wavelength the contribution of intensities from available rest-frame wavelengths is complete. This is necessary, for example, because setting the bluest observer-frame wavelength to the bluest rest-frame wavelength would mean that red shifted intensities from even bluer missing rest-frame wavelengths would not be included.

Next, the intensity at each surface element is interpolated onto the common observer-frame grid for both stars. Once this is done, the sum of the surface integrals (Linnell 1984, see equation 35) over both stars yield the flux at the Earth at each observer-frame wavelength

$$F_{1,2}(\lambda_{\mathrm{obs}}) = \frac{a^2}{d^2}\int_0^{\pi}\int_0^{2\pi} I_{1,2}(\lambda_{\mathrm{obs}}, \vartheta, \varphi) r_{1,2}^2(\varphi, \vartheta)\sin\vartheta \frac{\mu_{1,2}(\varphi, \vartheta)}{\cos\beta_{1,2}(\varphi, \vartheta)}\,\mathrm{d}\varphi\,\mathrm{d}\vartheta \tag{A67}$$

$$F_{\mathrm{total}}(\lambda_{\mathrm{obs}}) = F_1(\lambda_{\mathrm{obs}}) + F_2(\lambda_{\mathrm{obs}}) \tag{A68}$$

where d is distance and the radii $r_{1,2}$ are in units of the semi-major axis a, thus intensities with units erg cm^{-2} s^{-1} Å^{-1} sr^{-1}, yield fluxes with units of erg cm^{-2} s^{-1} Å^{-1}. This 2-D integral is computed using the IDL routine `INT_TABULATED_2D` (version 1.6) which first constructs a Delaunay triangulation of points in the $\varphi\vartheta$-plane. For computing the flux a sampling in φ and ϑ of 37×37 is sufficient for better than 1% flux accuracy (Aufdenberg et al. 2006).

Synthetic colors for photometric bands i and j are computed from the total flux

$$(i-j) = 2.5 \log \left\{ \frac{\int_0^\infty F_{\rm total}(\lambda_{\rm obs}) S_j(\lambda_{\rm obs})\, d\lambda_{\rm obs}}{\int_0^\infty F_{\rm total}(\lambda_{\rm obs}) S_i(\lambda_{\rm obs})\, d\lambda_{\rm obs}} \right\} \tag{A69}$$

where S_i and S_j are the filter sensitivity functions.

A.9. Testing Code with ι Peg

As a test of the `TRIAXIAL` code, we have computed models for a less complicated binary system, ι Peg (HR 8430 = HD 210027 = HIP 109176), in order to compare our best-fit parameters to those in the literature. Two snapshots of CHARA/MIRC four-telescope (S2-E2-W1-W2) triple and amplitude and closure phase data were taken JD 2453980.906 and JD 2453980.914 (2 September 2006 UT). Table 5 shows that our best-fit parameters to these data and 13-color photometry (Johnson & Mitchell 1975), $\chi_\nu = 1.3$, is in good agreement with the Boden et al. (1999) fit based two-telescope K-band interferometry from the Palomar Testbed Interferometer and optical spectroscopy.

Our fit to CHARA/MIRC H-band prism four-telescope resolves the ambiguity in longitude of the node, Ω.

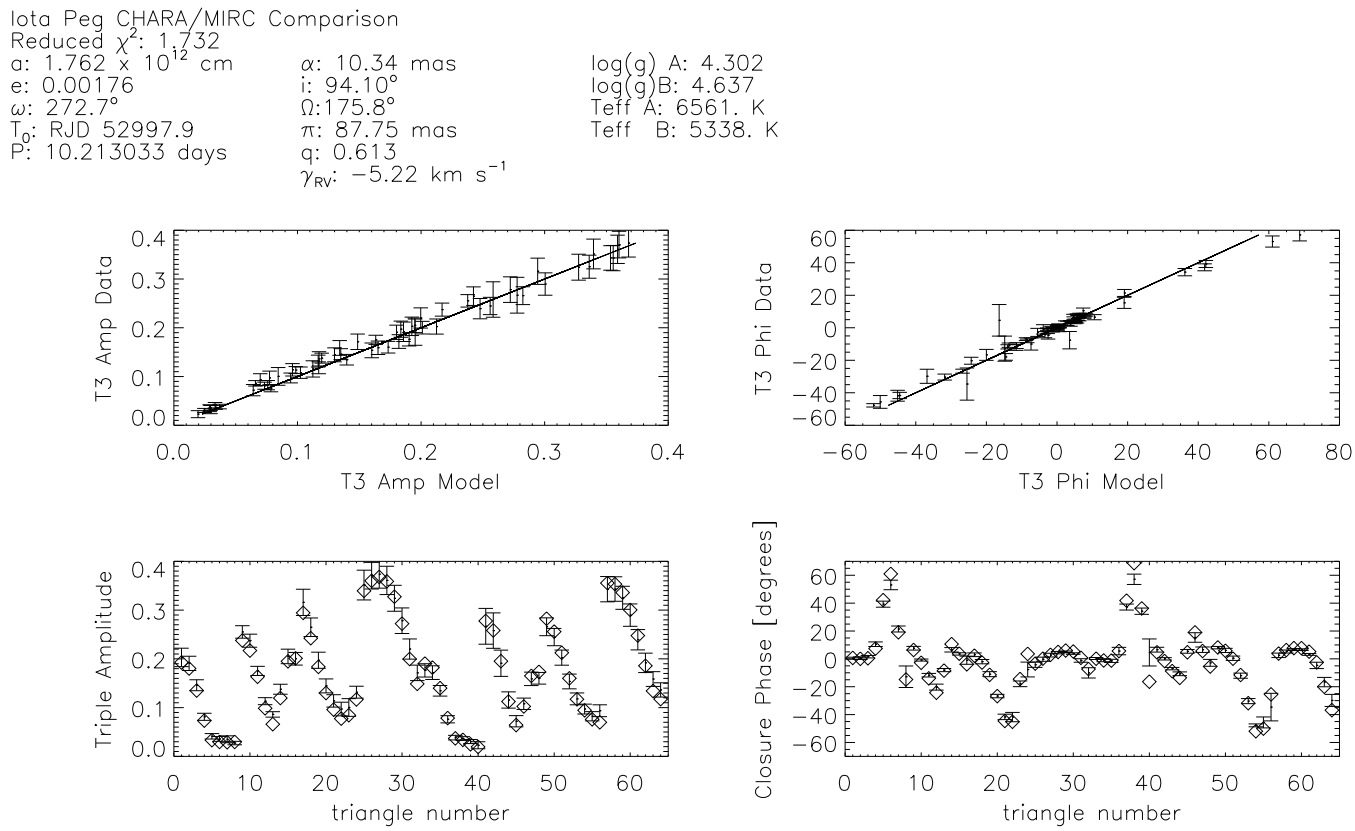

Fig. 11.— MIRC triple amplitude (T3 Amp) and closure phase (T3 Phi) data for ι Peg in comparison to the global best fit.

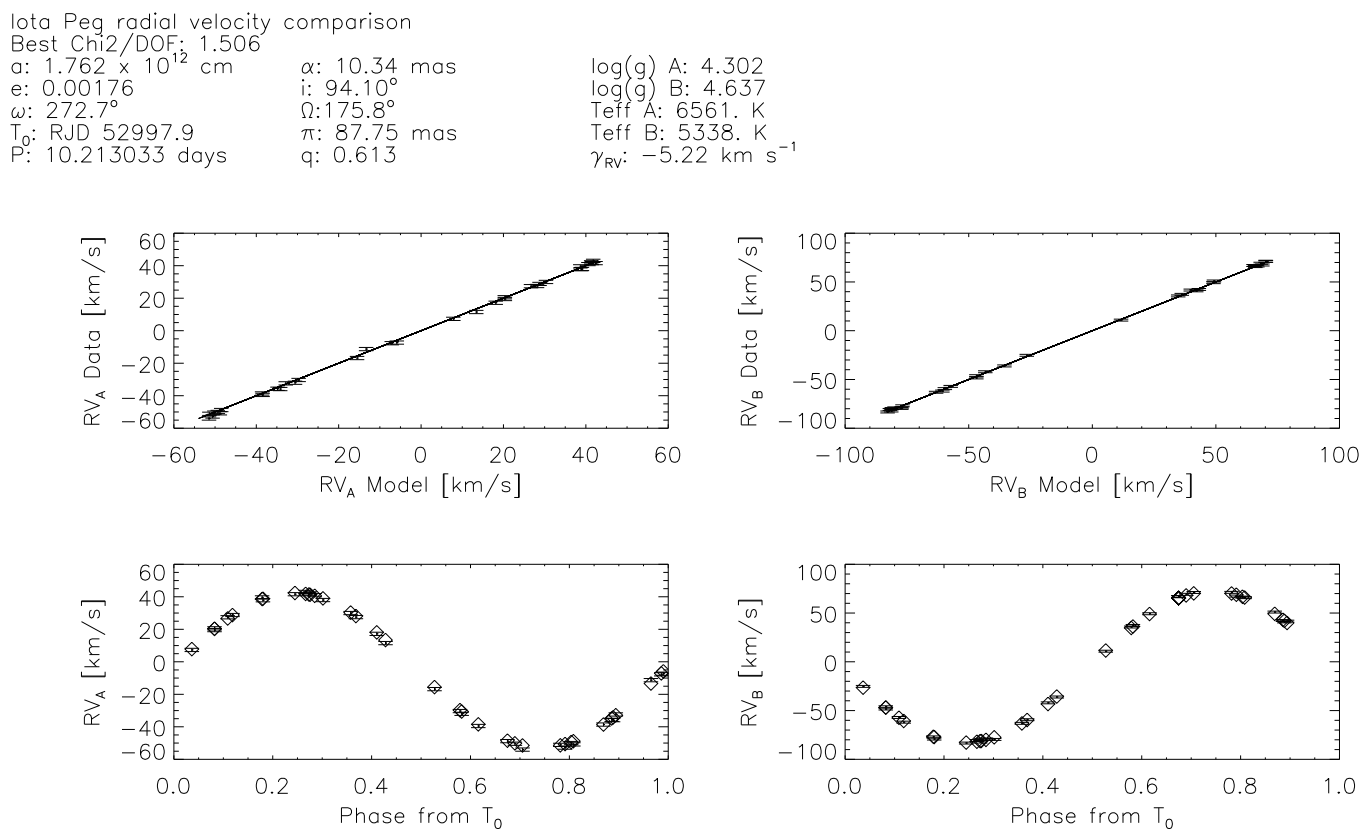

Fig. 12.— Archival radial velocity data for ι Peg in comparison to the global best fit.

A.10. Pseudo Code for TRIAXIAL

```
GET model intensity, direction cosine of emergent angle, wavelength, surface gravity,
    and effective temperature arrays from common block

CALL gauleg.pro (from Numerical Recipes) RETURNING Gaussian quadrature
     nodes and weights (for 2-D Fourier transform)

CALCULATE logarithm of intensity arrays (to speed up interpolation)

CASE interferometric dataset OF

        FLUOR : READ  Julian dates, (u,v) coordinates
        SUSI  : READ  Julian dates, (u,v) coordinates
        MIRC  : READ  Julian dates, (u,v) coordinates for each triangle, wavelengths

ENDCASE

CASE photometric dataset OF

        Hipparcos       : READ Julian dates
        Shobbrook       : READ Julian dates (Shobbrook et al. 1969 MNRAS  145, 131)
        Sterken         : READ Julian dates (Sterken et al. 1986 A&A 169, 166)

ENDCASE

CASE photometric dataset OF

Hipparcos : SET wavelength grid from 330 nm to 919 nm for Hipparcos filter
        Shobbrook       : SET wavelength grid from 470 nm to 710 nm for Johnson V-band
        Sterken         : SET wavelength grid from 313 nm to 594 nm to for Stroemgren uvby bands
        Johnson 13 color: SET wavelength grid from 316 nm to 1200 nm

ENDCASE

INIT array to save synthetic images of binary for each Julian date

IF high-resolution spectrum simulation THEN

        INIT wavelength and flux arrays to save synthetic data
SET Julian date
        SET wavelength grid

ENDIF

CASE interferometric dataset OF

FLUOR : GET number of single baselines: sets of (u,v), wavelength, Julian date
SUSI  : GET number of single baselines: sets of (u,v), wavelength, Julian date
        MIRC  : GET number of triangles:  sets of [(u1,v1), (u2,v2), (u3,v3)], wavelength, Julian date
```

```
ENDCASE

CASE interferometric dataset OF

        FLUOR : INIT spatial frequency, squared visibility,and visibility normalization arrays
        SUSI  : INIT spatial frequency, squared visibility,and visibility normalization arrays
        MIRC  : INIT visibility normalization, triple amplitude and closure phase arrays

ENDCASE

BEGIN READ binary star orbital and stellar parameters

        semi-major axis
        eccentricity
        longitude of periastron, little omega
        epoch of periastron
        period of apsidal motion
        inclination
        longitude of the ascending node, big omega
        parallax
        mass ratio
        surface gravity at the pole of Spica A
        surface gravity at the pole of Spica B
        effective temperature at the pole of Spica A
        effective temperature at the pole of Spica B
        asynchronous rotation factor for Spica A
        asynchronous rotation factor for Spica B

END READ

COMPUTE angular size of the semi-major axis

COMPUTE radius of at the pole of each star

COMPUTE radius of at the pole of each star in units of the semi-major axis

COMPUTE apsidal constant k2obs-bar (equation (2), page 492, from Claret and Gimenez 1993 A&A 277, 487)

PRINT all input and derived orbital and stellar parameters

BEGIN INIT of arrays to store parameters and orientations of stellar surface elements on each star

      array of co-latitudes, theta  (77 elements each for visibilities, 37 for fluxes)
      array of longitudes, phi (77 elements each for visibilities, 37 for fluxes)
      array of x coordinates set for each surface element
           (5929 points in the 77x77 grid or 1369 points in the 37x37 grid)
      array of y coordinates set for each surface element
      array of z coordinates set for each surface element
      array of direction cosine between the surface normal and the line of sight, mu, at each element
      array of the angle between the radius vector and the surface normal at each element
```

```
      array of interpolated intensities at each element
      array of projected velocities at each element
      array of observed wavelengths Doppler shifted from the non-orbiting, non-rotating frame
      array of the stellar radius for each element
      array of radii at four points on each star (pole, point, back, side)
      array of surface gravity at four points on each star (pole, point, back, side)
      array of temperature at four points on each star (pole, point, back, side)
      array of velocity at four points on each star (pole, point, back, side)

END INIT

FOR each Julian date (and multiple MIRC triangles for the same Julian date)

     IF first Julian date THEN

          SET build image to false

     ELSE

          IF same Julian date as last triangle THEN

                  SET build image to false

          ELSE

                  SET build image to true

          ENDIF

     ENDIF first Julian date

     IF build image is true THEN

          COMPUTE orbital phase for this Julian date

          CALL 'orbit2' with orbital elements and Julian date RETURNING angular separation of components,
               position angle,true anomaly, longitude of periastron for this Julian date

          COMPUTE the angle of the Spica B from the line of nodes (psi)

          COMPUTE the direction cosines for the line of sight l_0, m_0, n_0

          FOR each star

               FOR each longitude

                    FOR each co-latitude starting at the pole

                         COMPUTE direction cosines of the radius vector at this
                              co-latitude, longitude point
```

```
IF at the pole of the star THEN

     COMPUTE the polar radius from the polar surface gravity, semi-major axis,
          period, and mass ratio

     CALL 'big_omega' RETURNING the modified potential at the pole
          (CALL instead 'big_omega2' for secondary)

     SET the surface potential this modified potential value

ELSE

     SET the guess radius for this co-latitude, longitude point based on the
          radius at the last co-latitude

     CALL 'big_omega' RETURNING the modified potential for the guess radius
          (CALL instead 'big_omega2' for secondary)

     CALL 'deriv_big_omega' RETURNING the derivative of the modified potential
          w.r.t radius (CALL instead 'deriv_big_omega2' for secondary)

     COMPUTE new guess radius = old guess radius - (modified potential for
          the guess radius - surface potential)/potential derivative

     COMPUTE difference between these new and old guess radii

     WHILE this difference is greater than 1 part in 1 million

          SET guess radius to new guess radius

          CALL 'big_omega' RETURNING the modified potential for the guess radius
              (CALL instead 'big_omega2' for secondary)

          CALL 'deriv_big_omega' RETURNING the derivative of the modified
               potential w.r.t radius(CALL instead 'deriv_big_omega2' for seconar

          COMPUTE a new guess radius = old guess radius - (modified potential
               for the guess radius - surface pontential)/potential derivative

          COMPUTE difference between these new and old guess radii

     ENDWHILE

ENDIF at the pole of the star

SET radius to the final guess radius

COMPUTE the x, y, z coordinates
```

```
                    COMPUTE projected velocity for the line-of-sight

                    CALL 'deriv_omega_x' returning the x-component of the gravitational acceleration
                                        (CALL instead 'deriv_omega_x_2' for secondary)

                    CALL 'deriv_omega_y' returning the y-component of the gravitational acceleration

                    CALL 'deriv_omega_z' returning the z-component of the gravitational acceleration

                    COMPUTE effective gravitational acceleration from the components

                    COMPUTE effective temperature from Von Zeipel's pure radiative law

                    COMPUTE the direction cosines for the normal of the surface element

                    COMPUTE the angle between the radius vector and the surface normal

                    COMPUTE the angle between the surface normal and the line-of-sight

                    IF this surface area is visible (mu > 0) THEN

                         FOR all rest-frame wavelengths

                              CALL 'get_intensity' RETURNING the interpolated intensity at this
                                   gravity, temperature, and emergent angle

                              COMPUTE the observed Doppler-shifted wavelength which goes
                                   with this intensity

                          ENDFOR

                     ELSE
                          surface element not visible, do nothing

                     ENDIF surface area visible

                ENDFOR co-latitudes

            ENDFOR longitudes

       ENDFOR star

  ENDIF build image

  IF photometric or high resolution spectrum simulation THEN

      COMPUTE the shortest observed wavelength common to all visible surface elements

      COMPUTE the longest observed wavelength common to all visible surface elements
```

```
    SET observed wavelength range between these two wavelengths

    INIT intensity array to store only the visible surface elements at each observed wavelength

    INIT array to store the total flux of both stars at each observed wavelength


    FOR each surface element

         FOR each star

              COMPUTE the interpolated intensity on a regular wavelength grid from
                   irregular Doppler-shifted wavelength grid

         ENDFOR

     ENDFOR

     COMPUTE the wavelength-independent parts of integrand of the 2-D surface integral for the flux

     FOR all observed wavelengths

          FOR each star

               COMPUTE the flux integrand

               CALL 'my_int_tabulated_2D' RETURNING the flux at this wavelength

          ENDFOR

          SET flux of each star, the flux ratio and the total flux

     ENDFOR

     IF photometric simulation THEN

          CASE photometric dataset OF

               Hipparcos : CALL 'get_band_flux' RETURNING H-magnitude for this Julian date
               Shobbrook : CALL 'get_band_flux' RETURNING V-magnitude for this Julian date
               Sterken   : CALL 'get_band_flux' RETURNING uvby-magnitudes for this Julian date
               Johnson 13 color: CALL 'get_band_flux' RETURNING 13 magnitudes for this Julian date

          ENDCASE

     ENDIF photometric simlution

ENDIF photometric or high resolution spectrum simulation

IF interferometric simulation THEN
```

```
COMPUTE tranformation of x,y,z coordinates into the plane of the sky

CALL 'triangulate' RETURNING the Delaunay triangulation

FOR all rest wavelengths

     FOR each star

          CALL 'trigrid' RETURNING a 513x513 interpolated image

     ENDFOR

     COMPUTE the sum of both images for an image of the binary

     IF MIRC dataset THEN

          INIT array to store visibility for each leg of the triangle

          INIT array to store phase for each leg of the the triangle

          CALL 'my_int_2d' RETURNING the normalization for the visibility at
               zero spatial frequency

          FOR each leg of the triangle

               GET (u,v) coordinates for this Julian date

               COMPUTE the corresponding spatial frequencies at the current wavelength

               CALL 'my_int_2d' RETURNING the real (cosine) of the Fourier transform

               CALL 'my_int_2d' RETURNING the imaginary (sine) part of the Fourier transform

               COMPUTE the visibility as the square root of (real part^2 + imaginary part^2)

               CALL 'atan' RETURNING the phase

          ENDFOR each leg of the triangle

          COMPUTE the triple amplitude as (vis(leg 1)*vis(leg 2)*vis(leg 3))/normalization^3

          COMPUTE the closure phase as the sum of the phases on each leg

          WHILE closure phase > 180  degrees DO closure phase = closure phase - 360 degrees

          WHILE closure phase < -180 degrees DO Closure phase = closure phase + 360 degrees

     ELSE
```

```
                    GET (u,v) coordinates for this Julian date

                    COMPUTE the corresponding spatial frequencies at the current wavelength

                    CALL 'my_int_2d' RETURNING the real (cosine) of the Fourier transform

                    CALL 'my_int_2d' RETURNING the imaginary (sine) part of the Fourier transform

                    COMPUTE the squared visibility as absolute value of (real part^2 + imaginary part^2)

              ENDIF MIRC dataset

         ENDFOR all rest wavelengths

    ENDIF inteferometric simulation

ENDFOR each Julian date
```

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This 2-column preprint was prepared with the AAS LaTeX macros v5.2.

Table 1
Archival Radial Velocity Data

RJD days	Spica A km/s	Spica B km/s
Baker (1910)		
17686.675	+107.0	–132.0
17686.720	+89.0	–159.0
17700.646	–80.0	···
17700.693	–71.0	···
17714.689	+76.0	–133.0

Note.—Table 1 is published in its entirety in the electronic edition of the Astrophysical Journal. A portion is shown here for guidance regarding its form and content.

Table 2

Orbital and Stellar Parameters for Spica based on fit to Interferometry and Spectrophotometry (MUST BE REVISED)

Parameter	Primary Component	Secondary Component
Fixed Parameters		
Period (d)	4.0145898	
T_0 (HJD)	2440678.085	
$v \sin i$ (km/s)	161±2	70±5
$\frac{\omega_{\rm rot}}{\omega_{\rm orb}}$	1.80	1.80
Adjusted Parameters		
Mass ratio, $q = M_2/M_1$	0.628±?	
$T_{\rm eff}^{\rm mean}$ (K)	24025±?	23127±?
$\log(g)^{\rm pole}$ (cm s^{-2})	3.74±?	4.23±?
$\pi_{\rm orb}$ (mas)	12.8±?	
a (10^{12} cm)	1.853±?	
i (deg)	115.7±?	
e	0.119± ?	
ω at T_0 (deg)	133.8±?	
Ω (deg)	308.9±?	
Apsidal Period, U (years)	110±?	
β	0.114±?	
Derived Parameters		
a (mas)	1.586±?	
Mass ($M_\odot$)	9.66±?	6.07±?
Radius$_{\rm pole}$ ($R_\odot$)	6.94±?	4.23±?

Table 3

α Vir Best Fit Parameters Compared to the Riddle Spectroscopic Parameters

Parameter	Our Solution		Riddle (2000)	
	Primary Component	Secondary Component	Primary Component	Secondary Component
Mass ratio, $q = M_2/M_1$	0.68±0.04		0.636±0.029	
$M_{1,2} \sin^3 i$ ($M_\odot$)	5.43±1.29	3.69±1.26	7.64±0.38	4.86±0.63
$a_{1,2} \sin i$ ($R_\odot$)	8.99±0.76	13.2±1.1	9.61±0.17	15.11±0.66

Table 4

α Vir Best Parameters Compared to the Herbison-Evans et al. (1971) Solution

Parameter	Our Solution		Herbison-Evans et al. (1971)	
	Primary Component	Secondary Component	Primary Component	Secondary Component
Mass ($M_\odot$)	9.66±?	6.07±?	10.9±0.9	6.8±0.7
$\pi_{\rm orb}$ (mas)	12.80±?		11.90±0.57	
a (10^{12} cm)	1.85±?		1.93±0.06	
e	0.118±?		0.146 (fixed)	
i (deg)	115.7±?		65.9±1.8	
Ω (deg)	308.9±?		131.6±2.1	
Radius$_{\rm pole}$ ($R_\odot$)	6.9±?	3.1±?	8.1±0.5	...
$T_{\rm eff}^{\rm mean}$ (K)	24025±?	23147±?	22400± 1000	...
$\log(g)$ (cm s^{-2})	3.74±?	4.23±?	3.7±0.1	...
M_2/M_1, q	0.628±?		0.624±0.082	
a (mas)	1.59 ±?		1.54±0.05	
ω at T_0 (deg)	133.8±?		138±15	
Apsidal Period, U (years)	110±?		124± 11	

Table 5

Orbital and Physical Parameters for ι Peg

Parameter	Primary Component	Secondary Component	Comment	PTI 1999 Value
a (10^{12} cm)	1.764±0.042			
a (mas)	10.35 ±0.35		from a and $\pi_{\rm orb}$	10.33±0.10
i (deg)	94.99±0.41			95.67±0.22
Ω (deg)	274.53±0.23			94.09±0.23±180
$\pi_{\rm orb}$ (mas)	87.79±2.12			86.91±1.0
System Distance (pc)	11.39±0.28		from $\pi_{\rm orb}$	11.51±0.13
$T_{\rm eff}$ (K)	6520±35	5375±158		F5V & G8 V
$\log(g)$ (cm s^{-2})	4.31±0.01	4.64±0.10		
Mass ratio, q	$M_B/M_A = 0.617$		fixed	0.6175±0.0035 from K_A & K_B
Period (d)	10.213		fixed	
e	0		fixed	
ω (deg)	0		fixed	
T_0 (HJD)	24450661.559		fixed	
Mass ($M_\odot$)	1.30±0.12	0.80±0.07	from a, P, q	1.32±0.06 & 0.819±0.009
Radius ($R_\odot$)	1.32±0.06	1.03±0.12	from $\log(g)$, Mass	

Table 6
CHARA/FLUOR K'-band Spica Measurements

No.	Julian Date	Telescope Pair	u (meters)	v (meters)	Projected Baseline (meters)	Position Angle (degrees)	V^2 $\times$ 100	$\sigma V^2_{\rm total}$ $\times$ 100	Calibration Star(s) HD Number
1	2453858.79736	E1-W1	289.896	57.677	295.578	78.747	16.232	1.090	118098
2	2453858.83004	E1-W1	265.075	46.577	269.135	80.034	18.245	1.147	118098
3	2453863.77701	E1-W1	293.494	60.060	299.576	78.434	30.007	2.300	118098
4	2453863.80500	E1-W1	275.063	50.331	279.630	79.630	25.642	2.004	118098
5	2453863.82360	E1-W1	258.051	44.278	261.822	80.263	41.791	1.410	118098
6	2453864.75001	E1-W1	302.122	68.892	309.877	77.154	39.759	3.572	97603, 118098
7	2453864.77155	E1-W1	294.812	61.040	301.065	78.302	30.076	3.297	118098
8	2453864.79925	E1-W1	277.469	51.346	282.180	79.515	27.045	3.554	118098
9	2453865.76143	E1-W1	297.938	63.710	304.674	77.929	20.826	0.896	108767, 118098
10	2453865.77954	E1-W1	289.144	57.222	294.752	78.805	19.856	0.869	118098
11	2453865.79447	E1-W1	279.047	52.041	283.858	79.435	19.673	0.846	118098

Table 7
SUSI 700 nm Spica Measurements

No.	Julian Date	u (meters)	v (meters)	Projected Baseline (meters)	Position Angle (degrees)	V^2 $\times$ 100	$\sigma V^2_{\rm total}$ $\times$ 100
1	2453129.02126	0.82750	−37.78125	37.79031	−1.25471	78.902	2.048
2	2453129.03145	0.46907	37.78348	37.78639	0.71127	80.232	2.646
3	2453129.04179	1.78177	37.76929	37.81129	2.70093	78.240	4.017
4	2453129.05466	3.40504	37.72856	37.88190	5.15702	81.607	3.862
5	2453129.06211	4.33462	37.69342	37.94183	6.56001	81.580	2.612
6	2453129.07112	5.44745	37.63962	38.03177	8.23503	91.357	3.587
7	2453129.07846	6.34064	37.58684	38.11790	9.57525	95.558	3.529
8	2453129.09067	7.79441	37.48161	38.28347	11.74739	94.621	3.651
9	2453129.10717	9.68593	37.30551	38.54243	14.55478	84.782	3.600
10	2453129.11475	10.52046	37.21213	38.67070	15.78644	86.508	5.951
11	2453129.98278	5.31240	−37.64684	38.01982	−8.03205	65.204	2.142
12	2453129.99978	3.19921	−37.73517	37.87054	−4.84598	63.340	2.641
13	2453130.00737	2.24314	−37.76034	37.82691	−3.39964	62.477	2.616
14	2453130.01472	1.31129	−37.77629	37.79904	−1.98805	64.897	3.152
15	2453130.02317	0.23780	−37.78426	37.78501	−0.36059	65.678	3.195
16	2453130.03484	1.24629	37.77708	37.79763	1.88953	60.045	2.013
17	2453130.04292	2.27039	37.75975	37.82795	3.44089	65.275	4.136
18	2453130.05161	3.36490	37.72988	37.87963	5.09638	64.082	4.228
19	2453130.05919	4.31107	37.69442	37.94014	6.52450	58.498	2.965
20	2453130.06727	5.30947	37.64700	38.01956	8.02765	58.625	3.777
21	2453130.07764	6.56997	37.57186	38.14196	9.91869	55.954	3.463
22	2453130.08525	7.47854	37.50659	38.24490	11.27647	64.343	2.739
23	2453130.09306	8.39134	37.43105	38.36012	12.63572	57.725	2.510
24	2453130.11773	11.13372	37.13671	38.76977	16.68892	63.828	3.261
25	2453132.03955	2.53426	37.75363	37.83859	3.840290	68.732	3.164
26	2453132.04757	3.54294	37.72390	37.88991	5.36534	63.023	2.968
27	2453132.09915	9.70550	37.30344	38.54535	14.58371	67.859	2.289
28	2453141.99144	0.21139	−75.56896	75.56926	−0.16027	27.990	1.663
29	2453142.00103	2.22751	75.55717	75.59000	1.68865	24.070	1.440
30	2453142.01380	5.46174	75.49725	75.69456	4.13777	27.986	1.787
31	2453142.02535	8.35688	75.39987	75.86157	6.32451	25.821	1.474
32	2453144.93642	11.94466	−75.21932	76.16181	−9.02310	26.936	1.689
33	2453144.94824	9.04147	−75.37064	75.91101	−6.84051	31.396	2.004
34	2453144.95692	6.87659	−75.45491	75.76762	−5.20727	29.211	3.292
35	2453144.95940	6.25561	−75.47472	75.73352	−4.73804	28.250	3.268
36	2453144.97066	3.41188	−75.54112	75.61813	−2.58605	26.004	3.296
37	2453144.98157	0.63852	−75.56810	75.57079	−0.48411	28.551	2.881
38	2453144.99130	1.83519	75.56099	75.58327	1.39129	32.155	3.266
39	2453145.91826	15.63446	−74.95951	76.57261	−11.78139	24.543	2.728
40	2453145.92666	13.64109	−75.10963	76.33829	−10.29361	29.951	6.142
41	2453145.93912	10.61811	−75.29404	76.03905	−8.02702	23.405	4.800
42	2453145.95238	7.33140	−75.43918	75.79459	−5.55074	24.307	1.411
43	2453145.96090	5.19031	−75.50424	75.68243	−3.93243	24.884	1.635
44	2453145.97593	1.37910	−75.56451	75.57710	−1.04556	26.768	1.311
45	2453145.98784	1.65007	75.56254	75.58056	1.25097	30.862	1.508
46	2453145.99526	3.53507	75.53906	75.62173	2.67936	30.033	1.456
47	2453146.00456	5.88420	75.48565	75.71464	4.45726	32.571	2.414
48	2453146.01602	8.75095	75.38334	75.88957	6.62159	28.919	2.115
49	2453146.02662	11.36288	75.25327	76.10630	8.58652	27.221	1.099

Table 7—*Continued*

No.	Julian Date	u (meters)	v (meters)	Projected Baseline (meters)	Position Angle (degrees)	V^2 $\times$ 100	$\sigma V^2_{\rm total}$ $\times$ 100
50	2453184.91975	11.26871	75.25859	76.09756	8.51581	35.089	0.964
51	2453184.93392	14.68223	75.03416	76.45713	11.07140	49.867	1.647
52	2453184.94853	18.07771	74.74215	76.89729	13.59687	46.988	2.524
53	2453184.96119	20.89814	74.44106	77.31884	15.68123	52.273	2.093
54	2453184.97208	23.21801	74.14801	77.69816	17.38693	59.604	2.127
55	2453184.98404	25.64098	73.79133	78.11927	19.16129	69.972	3.327
56	2453186.88652	3.26100	56.64272	56.73652	3.29496	39.396	1.806
57	2453186.89601	5.05345	56.59463	56.81979	5.10252	44.155	2.077
58	2453186.90457	6.65707	56.53344	56.92405	6.71591	43.900	2.542
59	2453186.91536	8.64891	56.43270	57.09162	8.71338	42.253	2.442
60	2453186.92529	10.44613	56.31709	57.27771	10.50823	42.972	3.640
61	2453186.93451	12.07969	56.19036	57.47413	12.13267	51.377	4.176
62	2453186.94604	14.06373	56.00648	57.74525	14.09603	48.831	4.259
63	2453186.95623	15.75630	55.82108	58.00219	15.76248	66.725	3.778
64	2453206.83751	5.76000	75.48915	75.70858	4.36334	25.947	1.274
65	2453206.85125	9.19596	75.36371	75.92269	6.95690	28.410	1.464
66	2453206.86147	11.70529	75.23351	76.13866	8.84352	33.887	2.175
67	2453206.87339	14.57323	75.04236	76.44432	10.99004	31.615	2.087
68	2453206.88516	17.32459	74.81324	76.79298	13.03823	30.036	1.436
69	2453206.89922	20.48349	74.48894	77.25397	15.37555	32.950	1.723
70	2453206.93361	27.48750	73.47895	78.45202	20.51015	38.060	1.876
71	2453206.94586	29.68734	73.05159	78.85349	22.11626	34.603	2.127
72	2453207.85424	10.60564	75.29470	76.03796	8.01765	41.864	1.369
73	2453207.86367	12.90058	75.15950	76.25861	9.73949	45.894	2.014
74	2453207.87588	15.80306	74.94572	76.59372	11.90696	40.408	1.568
75	2453207.88699	18.36390	74.71414	76.93787	13.80892	37.572	1.050
76	2453207.89835	20.88974	74.44204	77.31752	15.67504	34.090	1.176
77	2453207.91118	23.61239	74.09369	77.76516	17.67618	29.578	1.293
78	2453207.92117	25.62672	73.79360	78.11674	19.15086	32.551	2.345
79	2453207.93619	28.46158	73.29782	78.62971	21.22121	34.082	2.073
80	2453207.95063	30.94708	72.77434	79.08114	23.03748	29.591	2.000
81	2453207.96166	32.67395	72.34632	79.38247	24.30548	33.127	2.334
82	2453435.11179	26.12381	−102.64372	105.91595	−14.27916	23.017	2.052
83	2453435.14918	13.96292	−103.56212	104.49917	−7.67868	29.381	3.371
84	2453435.16498	8.55176	−103.77925	104.13100	−4.71072	31.960	4.113
85	2453435.18067	3.09631	−103.89075	103.93688	−1.70711	38.144	3.860
86	2453435.19627	2.35820	103.89777	103.92453	1.30023	33.938	2.918
87	2453435.21193	7.81170	103.80058	104.09411	4.30378	32.752	4.362
88	2453435.22395	11.95135	103.65557	104.34229	6.57708	31.586	5.195
89	2453435.23491	15.66380	103.47098	104.64988	8.60827	40.021	5.573
90	2453435.25120	21.03803	103.10619	105.23063	11.53246	34.721	6.220
91	2453484.99384	20.22381	−103.16927	105.13277	−11.09080	13.934	1.127
92	2453485.00518	16.48034	−103.42318	104.72801	−9.05388	9.100	4.149
93	2453485.00807	15.50999	−103.47969	104.63558	−8.52428	13.583	4.111
94	2453485.03008	7.99180	−103.79557	104.10278	−4.40283	16.669	2.380
95	2453485.04154	4.00696	−103.87945	103.95670	−2.20898	21.335	2.167
96	2453485.06031	2.55598	103.89608	103.92751	1.40926	22.857	2.372
97	2453485.07085	6.23131	103.83958	104.02638	3.43414	23.601	2.602
98	2453485.08257	10.28476	103.72150	104.23016	5.66279	24.574	3.054

Table 7—*Continued*

No.	Julian Date	u (meters)	v (meters)	Projected Baseline (meters)	Position Angle (degrees)	V^2 $\times$ 100	$\sigma V^2_{\rm total}$ $\times$ 100
99	2453485.09296	13.83415	103.56856	104.48843	7.60823	25.631	4.141
100	2453485.10749	18.69046	103.28032	104.95789	10.25770	27.138	4.616
101	2453485.11932	22.53213	102.98283	105.41898	12.34154	26.626	3.866
102	2453486.94792	35.57377	−111.11881	116.67426	−17.75203	9.178	2.998
103	2453486.96069	31.51875	−111.64143	116.00535	−15.76550	14.458	2.329
104	2453486.98042	24.86086	−112.32064	115.03908	−12.48053	12.154	1.782
105	2453486.99598	19.33599	−112.74022	114.38635	−9.73206	9.767	1.506
106	2453487.00091	17.54333	−112.85112	114.20659	−8.83621	11.505	1.466
107	2453487.02130	9.96915	−113.19373	113.63188	−5.03314	14.857	1.767
108	2453487.03123	6.21300	−113.29177	113.46201	−3.13899	17.192	2.036

Table 8

CHARA/MIRC H-band Prism S1-E2-W1 Spica Measurements

No.	Julian Date	u_1 (meters)	v_1 (meters)	u_2 (meters)	v_2 (meters)	u_3 (meters)	v_3 (meters)	λ (nm)	Φ_{123} (degrees)	$\sigma\Phi_{123}$ (degrees)
1	2454232.71483	−142.89221	137.55291	99.45849	0.40982	43.43372	−137.96272	1774.51	−30.19791	0.71056
2	2454232.71483	−142.89221	137.55291	99.45849	0.40982	43.43372	−137.96272	1745.42	−29.20487	0.52061
3	2454232.71483	−142.89221	137.55291	99.45849	0.40982	43.43372	−137.96272	1712.08	−25.43560	0.41578
4	2454232.71483	−142.89221	137.55291	99.45849	0.40982	43.43372	−137.96272	1677.85	−24.45156	0.27320
5	2454232.71483	−142.89221	137.55291	99.45849	0.40982	43.43372	−137.96272	1642.73	−22.43929	0.34480
6	2454232.71483	−142.89221	137.55291	99.45849	0.40982	43.43372	−137.96272	1606.71	−19.10461	0.20778
7	2454232.71483	−142.89221	137.55291	99.45849	0.40982	43.43372	−137.96272	1569.79	−16.56107	0.53389
8	2454232.71483	−142.89221	137.55291	99.45849	0.40982	43.43372	−137.96272	1534.88	−13.94534	0.34900
9	2454232.71508	−143.14743	137.59570	99.52207	0.38005	43.62535	−137.97575	1774.51	−28.94308	0.52795
10	2454232.71508	−143.14743	137.59570	99.52207	0.38005	43.62535	−137.97575	1745.42	−28.90604	0.42044
11	2454232.71508	−143.14743	137.59570	99.52207	0.38005	43.62535	−137.97575	1712.08	−25.46468	0.61458
12	2454232.71508	−143.14743	137.59570	99.52207	0.38005	43.62535	−137.97575	1677.85	−24.77909	0.29262
13	2454232.71508	−143.14743	137.59570	99.52207	0.38005	43.62535	−137.97575	1642.73	−23.56282	0.45823
14	2454232.71508	−143.14743	137.59570	99.52207	0.38005	43.62535	−137.97575	1606.71	−19.38707	0.43554
15	2454232.71508	−143.14743	137.59570	99.52207	0.38005	43.62535	−137.97575	1569.79	−16.29436	0.36740
16	2454232.71508	−143.14743	137.59570	99.52207	0.38005	43.62535	−137.97575	1534.88	−12.69718	0.26813
17	2454232.71533	−143.40711	137.63939	99.58661	0.34969	43.82050	−137.98908	1774.51	−29.92052	0.46961
18	2454232.71533	−143.40711	137.63939	99.58661	0.34969	43.82050	−137.98908	1745.42	−28.43200	0.40455
19	2454232.71533	−143.40711	137.63939	99.58661	0.34969	43.82050	−137.98908	1712.08	−25.48365	0.54333
20	2454232.71533	−143.40711	137.63939	99.58661	0.34969	43.82050	−137.98908	1677.85	−24.27989	0.70957
21	2454232.71533	−143.40711	137.63939	99.58661	0.34969	43.82050	−137.98908	1642.73	−22.32809	0.43222
22	2454232.71533	−143.40711	137.63939	99.58661	0.34969	43.82050	−137.98908	1606.71	−18.96272	0.56257
23	2454232.71533	−143.40711	137.63939	99.58661	0.34969	43.82050	−137.98908	1569.79	−15.42102	0.61530
24	2454232.71533	−143.40711	137.63939	99.58661	0.34969	43.82050	−137.98908	1534.88	−12.31377	0.49350
25	2454232.71558	−143.66405	137.68275	99.65031	0.31960	44.01373	−138.00235	1774.51	−30.06531	0.54180
26	2454232.71558	−143.66405	137.68275	99.65031	0.31960	44.01373	−138.00235	1745.42	−28.27298	0.41502
27	2454232.71558	−143.66405	137.68275	99.65031	0.31960	44.01373	−138.00235	1712.08	−25.61969	0.31316
28	2454232.71558	−143.66405	137.68275	99.65031	0.31960	44.01373	−138.00235	1677.85	−24.54648	0.26767
29	2454232.71558	−143.66405	137.68275	99.65031	0.31960	44.01373	−138.00235	1642.73	−21.99747	0.18271
30	2454232.71558	−143.66405	137.68275	99.65031	0.31960	44.01373	−138.00235	1606.71	−18.68123	0.41199
31	2454232.71558	−143.66405	137.68275	99.65031	0.31960	44.01373	−138.00235	1569.79	−15.35634	0.51777
32	2454232.71558	−143.66405	137.68275	99.65031	0.31960	44.01373	−138.00235	1534.88	−12.21221	0.39735
33	2454232.72498	−153.15488	139.38548	101.88793	−0.83653	51.26695	−138.54896	1774.51	−26.10902	0.34138
34	2454232.72498	−153.15488	139.38548	101.88793	−0.83653	51.26695	−138.54896	1745.42	−23.31361	0.33440
35	2454232.72498	−153.15488	139.38548	101.88793	−0.83653	51.26695	−138.54896	1712.08	−22.77009	0.32632
36	2454232.72498	−153.15488	139.38548	101.88793	−0.83653	51.26695	−138.54896	1677.85	−19.30108	0.32450
37	2454232.72498	−153.15488	139.38548	101.88793	−0.83653	51.26695	−138.54896	1642.73	−16.00891	0.25446
38	2454232.72498	−153.15488	139.38548	101.88793	−0.83653	51.26695	−138.54896	1606.71	−13.04812	0.31031
39	2454232.72498	−153.15488	139.38548	101.88793	−0.83653	51.26695	−138.54896	1569.79	−9.58884	0.27104
40	2454232.72498	−153.15488	139.38548	101.88793	−0.83653	51.26695	−138.54896	1534.88	−5.21298	0.28824
41	2454232.72597	−154.12513	139.57134	102.10322	−0.95991	52.02191	−138.61143	1774.51	−25.18880	0.31193
42	2454232.72597	−154.12513	139.57134	102.10322	−0.95991	52.02191	−138.61143	1745.42	−23.04039	0.32667
43	2454232.72597	−154.12513	139.57134	102.10322	−0.95991	52.02191	−138.61143	1712.08	−22.02601	0.29489
44	2454232.72597	−154.12513	139.57134	102.10322	−0.95991	52.02191	−138.61143	1677.85	−18.70491	0.28454
45	2454232.72597	−154.12513	139.57134	102.10322	−0.95991	52.02191	−138.61143	1642.73	−14.96361	0.26593
46	2454232.72597	−154.12513	139.57134	102.10322	−0.95991	52.02191	−138.61143	1606.71	−12.29135	0.27635
47	2454232.72597	−154.12513	139.57134	102.10322	−0.95991	52.02191	−138.61143	1569.79	−8.60116	0.28352
48	2454232.72597	−154.12513	139.57134	102.10322	−0.95991	52.02191	−138.61143	1534.88	−4.73805	0.29048
49	2454232.72711	−155.23312	139.78657	102.34568	−1.10213	52.88745	−138.68444	1774.51	−24.78659	0.38138
50	2454232.72711	−155.23312	139.78657	102.34568	−1.10213	52.88745	−138.68444	1745.42	−22.77860	0.28203

Table 8—*Continued*

No.	Julian Date	u_1 (meters)	v_1 (meters)	u_2 (meters)	v_2 (meters)	u_3 (meters)	v_3 (meters)	λ (nm)	Φ_{123} (degrees)	$\sigma\Phi_{123}$ (degrees)
51	2454232.72711	−155.23312	139.78657	102.34568	−1.10213	52.88745	−138.68444	1712.08	−21.08840	0.33174
52	2454232.72711	−155.23312	139.78657	102.34568	−1.10213	52.88745	−138.68444	1677.85	−18.18286	0.27826
53	2454232.72711	−155.23312	139.78657	102.34568	−1.10213	52.88745	−138.68444	1642.73	−14.37177	0.23349
54	2454232.72711	−155.23312	139.78657	102.34568	−1.10213	52.88745	−138.68444	1606.71	−11.61387	0.31851
55	2454232.72711	−155.23312	139.78657	102.34568	−1.10213	52.88745	−138.68444	1569.79	−7.85687	0.29589
56	2454232.72711	−155.23312	139.78657	102.34568	−1.10213	52.88745	−138.68444	1534.88	−3.31241	0.31803
57	2454232.72825	−156.33480	140.00344	102.58347	−1.24480	53.75133	−138.75865	1774.51	−24.50094	0.46959
58	2454232.72825	−156.33480	140.00344	102.58347	−1.24480	53.75133	−138.75865	1745.42	−22.20166	0.37921
59	2454232.72825	−156.33480	140.00344	102.58347	−1.24480	53.75133	−138.75865	1712.08	−20.72597	0.38930
60	2454232.72825	−156.33480	140.00344	102.58347	−1.24480	53.75133	−138.75865	1677.85	−17.14208	0.28418
61	2454232.72825	−156.33480	140.00344	102.58347	−1.24480	53.75133	−138.75865	1642.73	−13.40722	0.28250
62	2454232.72825	−156.33480	140.00344	102.58347	−1.24480	53.75133	−138.75865	1606.71	−10.97715	0.31946
63	2454232.72825	−156.33480	140.00344	102.58347	−1.24480	53.75133	−138.75865	1569.79	−6.61095	0.26611
64	2454232.72825	−156.33480	140.00344	102.58347	−1.24480	53.75133	−138.75865	1534.88	−3.71953	0.36328
65	2454232.72925	−157.28615	140.19336	102.78581	−1.36916	54.50034	−138.82420	1774.51	−23.51965	0.36144
66	2454232.72925	−157.28615	140.19336	102.78581	−1.36916	54.50034	−138.82420	1745.42	−22.09024	0.31491
67	2454232.72925	−157.28615	140.19336	102.78581	−1.36916	54.50034	−138.82420	1712.08	−19.96949	0.38118
68	2454232.72925	−157.28615	140.19336	102.78581	−1.36916	54.50034	−138.82420	1677.85	−16.49594	0.27019
69	2454232.72925	−157.28615	140.19336	102.78581	−1.36916	54.50034	−138.82420	1642.73	−13.18760	0.30448
70	2454232.72925	−157.28615	140.19336	102.78581	−1.36916	54.50034	−138.82420	1606.71	−10.06884	0.30348
71	2454232.72925	−157.28615	140.19336	102.78581	−1.36916	54.50034	−138.82420	1569.79	−5.85013	0.37795
72	2454232.72925	−157.28615	140.19336	102.78581	−1.36916	54.50034	−138.82420	1534.88	−3.56419	0.35379